

(1) $\frac{1}{n} \sum_{i=1}^n x_i^2$ (2) $\frac{1}{n} \sum_{i=1}^n x_i$ (3) $\frac{1}{n} \sum_{i=1}^n x_i^2$
 (4) $\frac{1}{n} \sum_{i=1}^n x_i$ (5) $\frac{1}{n} \sum_{i=1}^n x_i^2$ (6) $\frac{1}{n} \sum_{i=1}^n x_i$
 (7) $\frac{1}{n} \sum_{i=1}^n x_i^2$ (8) $\frac{1}{n} \sum_{i=1}^n x_i$ (9) $\frac{1}{n} \sum_{i=1}^n x_i^2$
 (10) $\frac{1}{n} \sum_{i=1}^n x_i$ (11) $\frac{1}{n} \sum_{i=1}^n x_i^2$ (12) $\frac{1}{n} \sum_{i=1}^n x_i$

MULTIVARIATE ANALYSIS.

06/01/2024

1D MULTIVARIATE NORMAL DISTRIBUTION

UNIVARIATE

X: Height



Random sample: $(x_1, x_2, \dots, x_n)$

MULTIVARIATE

X: Height

Y: Weight

Z: Age

$$\underline{x} = \begin{pmatrix} x \\ y \\ z \end{pmatrix}$$

Random sample: $(\underline{x}_1, \underline{x}_2, \dots, \underline{x}_n)$

UNIVARIATE NORMAL DISTRIBUTION

$$X \sim N(\mu, \sigma^2)$$

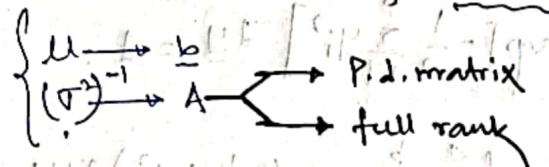
$$f(x) = \frac{1}{\sigma \sqrt{2\pi}} \exp\left[-\frac{1}{2} \left(\frac{x-\mu}{\sigma}\right)^2\right], \quad x \in \mathbb{R}, \mu \in \mathbb{R}, \sigma^2 > 0$$

$\underbrace{\frac{1}{\sigma \sqrt{2\pi}}}_{\text{constant}}$
 $\underbrace{\left(\frac{x-\mu}{\sigma}\right)^2}_{\text{Quadratic form}}$

Univariate: x, μ, σ^2

Multivariate: $\underline{x}, \underline{\mu}, \Sigma$

Now consider p variables $\underline{x} = (x_1, \dots, x_p)$.
 Let's construct the PDF of MVN distribution.



$$(\underline{x}-\underline{\mu})(\sigma^2)^{-1}(\underline{x}-\underline{\mu}) \rightarrow (\underline{x}-\underline{b})' \underline{A} (\underline{x}-\underline{b}) = \sum_i \sum_j a_{ij} (x_i - b_i)(x_j - b_j)$$

So, the PDF becomes -

$$f_{x_1, \dots, x_p}(x_1, \dots, x_p) = f_{\underline{x}}(\underline{x}) = k \cdot \exp\left[-\frac{1}{2} (\underline{x}-\underline{b})' \underline{A} (\underline{x}-\underline{b})\right]$$

From the properties of PDF -

$$k \int_{-\infty}^{\infty} \dots \int_{-\infty}^{\infty} \exp\left[-\frac{1}{2} (\underline{x}-\underline{b})' \underline{A} (\underline{x}-\underline{b})\right] d\underline{x} = 1$$

Since, A is a positive definite matrix, so $\exists$ a non-singular matrix C s.t.

$$C'AC = I.$$

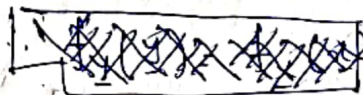
$$A = (CC')^{-1} \text{ or, } A^{-1} = CC'$$

Transformation: $\underline{x} - \underline{b} = C\underline{y}$

$$\underline{y} = \begin{pmatrix} y_1 \\ y_2 \\ \vdots \\ y_p \end{pmatrix}$$

$$(\underline{x} - \underline{b})' A (\underline{x} - \underline{b}) = \underline{y}' C' A C \underline{y} = \underline{y}' \underline{y}$$

Jacobian of Transformation $|J| = |C|$



$$f_Y(\underline{y}) = K \exp \left[-\frac{1}{2} \underline{y}' \underline{y} \right] |C|$$

$$= K |C|^{1/2} \exp \left[-\frac{1}{2} \underline{y}' \underline{y} \right]$$

$$= K |A^{-1}|^{1/2} \exp \left[-\frac{1}{2} \underline{y}' \underline{y} \right]$$

$$\int_{\mathbb{R}^p} f_Y(\underline{y}) d\underline{y} = 1$$

$$\Rightarrow K \int_{\mathbb{R}^p} |A^{-1}|^{1/2} \exp \left[-\frac{1}{2} \sum_{i=1}^p y_i^2 \right] d\underline{y} = 1$$

$$\Rightarrow K |A^{-1}|^{1/2} \int_{-\infty}^{\infty} \dots \int_{-\infty}^{\infty} \prod_{i=1}^p \exp \left(-\frac{1}{2} y_i^2 \right) dy_i = 1$$

$$\Rightarrow K |A^{-1}|^{1/2} \prod_{i=1}^p \int_{-\infty}^{\infty} \exp \left(-\frac{1}{2} y_i^2 \right) dy_i = 1$$

$$\Rightarrow K |A^{-1}|^{1/2} \left(\sqrt{2\pi} \right)^p = 1$$

$$\Rightarrow K = \frac{1}{(2\pi)^{p/2} |A^{-1}|^{1/2}}$$

$$E(y_i) = \int \dots \int y_i \cdot \frac{1}{\sqrt{2\pi}} \exp\left(-\frac{1}{2} y_j^2\right) dy_1 \dots dy_p$$

$$= \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} y_i \exp\left(-\frac{1}{2} y_i^2\right) dy_i \cdot \left(\prod_{j \neq i} \int_{-\infty}^{\infty} \frac{1}{\sqrt{2\pi}} \exp\left(-\frac{1}{2} y_j^2\right) dy_j \right)$$

$$= 0 \times 1 = 0$$

$$\therefore E(\underline{y}) = \underline{0} \quad \text{and} \quad E(\underline{x}) = \underline{b}$$

Therefore, $\underline{b} = \underline{\mu} = \begin{pmatrix} \mu_1 \\ \vdots \\ \mu_p \end{pmatrix}$

Now, $E(\underline{x} - \underline{b})(\underline{x} - \underline{b})' = c E(\underline{y} \underline{y}') c' = \underline{0}$

$$= c E \begin{pmatrix} y_1^2 & y_1 y_2 & \dots & y_1 y_p \\ y_2 y_1 & y_2^2 & \dots & y_2 y_p \\ \vdots & \vdots & \ddots & \vdots \\ y_p y_1 & y_p y_2 & \dots & y_p^2 \end{pmatrix} c'$$

Now, $E(y_i^2) = 1$ and $E(y_i y_j) = 0$ for $i \neq j$

$$= \left(\int_{-\infty}^{\infty} \frac{1}{\sqrt{2\pi}} y_i^2 \exp\left(-\frac{1}{2} y_i^2\right) dy_i \right) \cdot \left(\prod_{j \neq i} \int_{-\infty}^{\infty} \frac{1}{\sqrt{2\pi}} \exp\left(-\frac{1}{2} y_j^2\right) dy_j \right)$$

$$\therefore E(\underline{x} - \underline{b})(\underline{x} - \underline{b})' = c c' = A^{-1}$$

$$\Rightarrow E(\underline{x} - \underline{\mu})(\underline{x} - \underline{\mu})' = A^{-1}$$

$$\Rightarrow \Sigma = A^{-1}$$

$$\Rightarrow \boxed{A = \Sigma^{-1}}$$

Therefore, we finally get the PDF as -

$$f_{\underline{x}}(\underline{x}) = \frac{1}{(2\pi)^{p/2} |\Sigma|^{1/2}} \exp \left[-\frac{1}{2} (\underline{x} - \underline{\mu})' \Sigma^{-1} (\underline{x} - \underline{\mu}) \right]$$

$$\underline{x} \in \mathbb{R}^p$$

10/01/2024

Theorem > If the density of a p -dim RV $\underline{x}$ is

$$\frac{\sqrt{|A|}}{(2\pi)^{p/2}} \exp \left[-\frac{1}{2} (\underline{x} - \underline{b})' A (\underline{x} - \underline{b}) \right] \text{ then}$$

$$E(\underline{x}) = \underline{b} \quad \& \quad \text{cov}(\underline{x}) = A^{-1}$$

Conversely, given a $\underline{\mu}$ & Σ (p.d.) there is a

MVN distribution $\frac{1}{(2\pi)^{p/2}} \frac{|\Sigma|^{-1/2}}{\exp \left[-\frac{1}{2} (\underline{x} - \underline{\mu})' \Sigma^{-1} (\underline{x} - \underline{\mu}) \right]}$

$$\text{s.t. } E(\underline{x}) = \underline{\mu} \quad \& \quad \text{cov}(\underline{x}) = \Sigma$$

Remarks >

$$\underline{\theta} = (\theta_1, \dots, \theta_n)'$$

$f(\underline{\theta})$ is a real valued function of $\theta_1, \theta_2, \dots, \theta_n$

$$\frac{\partial f(\underline{\theta})}{\partial \underline{\theta}} = \begin{pmatrix} \partial f(\underline{\theta}) / \partial \theta_1 \\ \vdots \\ \partial f(\underline{\theta}) / \partial \theta_n \end{pmatrix}$$

e.g. If A is a $n \times n$ Symmetric matrix & $f(\underline{\theta}) = \underline{\theta}' A \underline{\theta}$

$$\frac{\partial f(\underline{\theta})}{\partial \underline{\theta}} = (A + A') \underline{\theta} = 2A \underline{\theta}$$

→ Characteristic function >

Univariate > $\phi(\theta) = E[\exp(i\theta X)] = \int e^{i\theta x} f(x) dx$

Multivariate > $\phi(\underline{\theta}) = E[\exp(i\underline{\theta}' \underline{x})] = \int_{\mathbb{R}^p} e^{i\underline{\theta}' \underline{x}} f(\underline{x}) d\underline{x}$

Properties > → $\phi(0) = 1$

→ $|\phi(\underline{\theta})| \leq 1$

→ Uniqueness Theorem >

Two RVs have the same characteristic function iff they have the same distribution.

INVERSION THEOREM

Given a characteristic function, we can find the PDF by inversion theorem,

Differentiating the characteristic fcn. w.r.t. $\underline{\theta}$ and then put $\underline{\theta} = \underline{0}$.

Characteristic function

MVN Distribution

$$\underline{X} \sim N_p(\underline{\mu}, \Sigma)$$

$$\phi(t) = E[\exp(it' \underline{X})]$$

$$\text{Let, } \underline{X} - \underline{\mu} = \underline{u}$$

$$\text{Then, } \phi(t) = E[\exp[it'(\underline{u} + \underline{\mu})]] = \exp(it' \underline{\mu}) E[\exp(it' \underline{u})]$$

$$= \exp(it' \underline{\mu}) E[\exp(i(t' \underline{e}) \underline{y})]$$

$$= \exp(it' \underline{\mu}) E[\exp(i \underline{u}' \underline{y})]$$

$$= \exp(it' \underline{\mu}) \prod_{j=1}^p E[\exp(i u_j y_j)]$$

$$[\underline{X} \sim N(\underline{\mu}, \sigma^2), \phi_X(t) = \exp[i \underline{\mu}' t - \frac{1}{2} \sigma^2 t^2]$$

$$\underline{\mu} = 0, \sigma^2 = 1, \phi_Z(t) = \exp(-\frac{1}{2} t^2)$$

$$\text{Treat } u_j = t, y_j = Z. \text{ So, } E(e^{i u_j y_j}) = e^{-\frac{1}{2} u_j^2}$$

$$\text{So, } \phi(t) = \exp(it' \underline{\mu}) \prod_{j=1}^p \exp(-\frac{1}{2} u_j^2)$$

$$= \exp(it' \underline{\mu}) \cdot \exp(-\frac{1}{2} \underline{u}' \underline{u})$$

$$= \exp(it' \underline{\mu} - \frac{1}{2} \underline{u}' \underline{u}) \quad [\underline{u}' \underline{u} = t' \underline{C} t]$$

$$= \exp(it' \underline{\mu} - \frac{1}{2} t' \Sigma t) \quad = t' \Sigma t]$$

DISTRIBUTION OF LINEAR COMBINATION

$$\underline{X} \sim N_p(\underline{\mu}, \Sigma)$$

$$\underline{Z} = A \underline{X}, \quad A \text{ is not necessarily full rank.}$$

$$\phi_z(t) = E[\exp(it'z)]$$

$$= E[\exp(it'A'x)]$$

$$= E[\exp(i\theta'z)]$$

$$= \exp[i\theta'\mu - \frac{1}{2}\theta'\Sigma\theta]$$

$$= \exp[it'(A\mu) - \frac{1}{2}t'(A\Sigma A')t]$$

Using uniqueness theorem, z follows M.V.N. with mean vector $A\mu$ & cov. matrix $A\Sigma A'$.

$$z = Ax \sim N_p(A\mu, A\Sigma A')$$

11/01/2024

$$x = (x_1, x_2, \dots, x_p)' = (x_1', x_2')'$$

$$x \sim N_p(\mu, \Sigma)$$

$$\mu = \begin{pmatrix} \mu_1 \\ \mu_2 \end{pmatrix} \rightarrow \begin{matrix} \gamma x_1 \\ (1-\gamma)x_1 \end{matrix}$$

$$\Sigma = \begin{pmatrix} \Sigma_{11} & \Sigma_{12} \\ \Sigma_{21} & \Sigma_{22} \end{pmatrix}$$

To find the distribution of x_1 & x_2 .

[Remark: Say x_1 & x_2 are independent.
then $\Sigma_{12} = 0 = \Sigma_{21}$

$$\text{Then } \Sigma = \begin{pmatrix} \Sigma_{11} & 0 \\ 0 & \Sigma_{22} \end{pmatrix}$$

$$\therefore \Sigma^{-1} = \begin{pmatrix} \Sigma_{11}^{-1} & 0 \\ 0 & \Sigma_{22}^{-1} \end{pmatrix} \text{ \& } |\Sigma| = |\Sigma_{11}| \cdot |\Sigma_{22}|$$

Now, the PDF of x is -

$$f_x(\mu, \Sigma) = \frac{1}{(2\pi)^{p/2} |\Sigma|^{1/2}} \exp\left[-\frac{1}{2}(\underline{x} - \underline{\mu})' \Sigma^{-1} (\underline{x} - \underline{\mu})\right]$$

$$\theta = [(\underline{x}_1 - \underline{\mu}_1)', (\underline{x}_2 - \underline{\mu}_2)']' \begin{pmatrix} \Sigma_{11}^{-1} & 0 \\ 0 & \Sigma_{22}^{-1} \end{pmatrix} \begin{pmatrix} \underline{x}_1 - \underline{\mu}_1 \\ \underline{x}_2 - \underline{\mu}_2 \end{pmatrix}$$

$$= (\underline{x}_1 - \underline{\mu}_1)' \Sigma_{11}^{-1} (\underline{x}_1 - \underline{\mu}_1) + (\underline{x}_2 - \underline{\mu}_2)' \Sigma_{22}^{-1} (\underline{x}_2 - \underline{\mu}_2).$$

$$= Q_1 + Q_2.$$

$$f_{\underline{x}}(\underline{\mu}, \Sigma) = \frac{1}{(2\pi)^{q/2} |\Sigma_{11}|^{1/2}} \exp\left(-\frac{Q_1}{2}\right) \cdot \frac{1}{(2\pi)^{\frac{p-q}{2}} |\Sigma_{22}|^{1/2}} \exp\left(-\frac{Q_2}{2}\right).$$

$$= f_{\underline{x}_1}(\underline{\mu}_1, \Sigma_{11}) \cdot f_{\underline{x}_2}(\underline{\mu}_2, \Sigma_{22})$$

i.e. $\underline{x}_1 \sim N_q(\underline{\mu}_1, \Sigma_{11})$ & $\underline{x}_2 \sim N_{p-q}(\underline{\mu}_2, \Sigma_{22})$ independently

General case ($\Sigma_{12} \neq 0$)

$$\text{Choose, } \underline{y}_1 = \underline{x}_1 + B \underline{x}_2$$

$$\underline{y}_2 = \underline{x}_2$$

Select B s.t. $\underline{y}_1$ & $\underline{y}_2$ are uncorrelated.

$$\text{i.e. } E[(\underline{y}_1 - E(\underline{y}_1))(\underline{y}_2 - E(\underline{y}_2))'] = 0$$

$$E[(\underline{x}_1 - E(\underline{x}_1)) + B(\underline{x}_2 - E(\underline{x}_2))](\underline{x}_2 - E(\underline{x}_2))' = 0$$

$$\Rightarrow \Sigma_{12} + B \Sigma_{22} = 0$$

$$\Rightarrow B = -\Sigma_{12} \Sigma_{22}^{-1}$$

$$\therefore \underline{y} = \begin{pmatrix} \underline{y}_1 \\ \underline{y}_2 \end{pmatrix} = \begin{pmatrix} \mathbf{I} & -\Sigma_{12} \Sigma_{22}^{-1} \\ 0 & \mathbf{I} \end{pmatrix} \begin{pmatrix} \underline{x}_1 \\ \underline{x}_2 \end{pmatrix} \quad [\text{Non-singular transformation}]$$

$$E \begin{pmatrix} \underline{y}_1 \\ \underline{y}_2 \end{pmatrix} = \begin{pmatrix} \mathbf{I} & -\Sigma_{12} \Sigma_{22}^{-1} \\ 0 & \mathbf{I} \end{pmatrix} \begin{pmatrix} \underline{\mu}_1 \\ \underline{\mu}_2 \end{pmatrix} = \begin{pmatrix} \underline{\mu}_1 - \Sigma_{12} \Sigma_{22}^{-1} \underline{\mu}_2 \\ \underline{\mu}_2 \end{pmatrix} = \begin{pmatrix} \underline{\nu}_1 \\ \underline{\nu}_2 \end{pmatrix}$$

$$\begin{aligned} \text{cov}(\underline{y}) &= E \begin{pmatrix} (\underline{y}_1 - \underline{\nu}_1)(\underline{y}_1 - \underline{\nu}_1)' & (\underline{y}_1 - \underline{\nu}_1)(\underline{y}_2 - \underline{\nu}_2)' \\ (\underline{y}_2 - \underline{\nu}_2)(\underline{y}_1 - \underline{\nu}_1)' & (\underline{y}_2 - \underline{\nu}_2)(\underline{y}_2 - \underline{\nu}_2)' \end{pmatrix} \\ &= \begin{pmatrix} \Sigma_{11} - \Sigma_{12} \Sigma_{22}^{-1} \Sigma_{21} & 0 \\ 0 & \Sigma_{22} \end{pmatrix} = \begin{pmatrix} \Sigma_{11.2} & 0 \\ 0 & \Sigma_{22} \end{pmatrix}. \end{aligned}$$

$$\underline{y}_1 \sim N(\underline{\mu}_1, \Sigma_{11.2})$$

$$\underline{y}_2 \sim N(\underline{\mu}_2, \Sigma_{22}) \equiv N(\underline{\mu}_2, \Sigma_{22})$$

$\underline{x}_2$

Now, Find distⁿ of $\underline{x}_1 = \underline{y}_1 + \Sigma_{12} \Sigma_{22}^{-1} \underline{y}_2$ [$\underline{y}_1, \underline{y}_2$ are independent]

15/01/2024

CONDITIONAL DISTRIBUTIONS

We have chosen B in such a way that

$$\underline{y}_1 = \underline{x}_1 + B \underline{x}_2$$

$$\Delta. \underline{y}_2 = \underline{x}_2$$

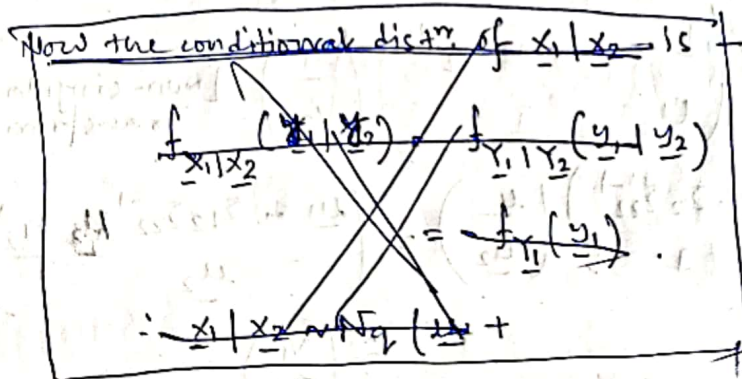
becomes independent.

and the PDF is given by -

$$f_{\underline{y}_1, \underline{y}_2}(\underline{y}_1, \underline{y}_2) = f_{\underline{x}_1, \underline{x}_2}(\underline{x}_1, \underline{x}_2) |J| \quad \text{with } |J| = 1$$

$$= f_{\underline{y}_1}(\underline{y}_1) \cdot f_{\underline{y}_2}(\underline{y}_2)$$

$$\text{where } \begin{cases} \underline{y}_1 \sim N(\underline{\mu}_1 - \Sigma_{12} \Sigma_{22}^{-1} \underline{\mu}_2, \Sigma_{11.2}) \\ \underline{y}_2 \sim N(\underline{\mu}_2, \Sigma_{22}) \end{cases}$$



Now,

$$f_{\underline{x}_1 | \underline{x}_2}(\underline{x}_1 | \underline{x}_2) = f_{\underline{y}_1}(\underline{y}_1)$$

$$\underline{y}_1 | \underline{x}_2 \sim N(\underline{\mu}_1 - \Sigma_{12} \Sigma_{22}^{-1} \underline{x}_2, \Sigma_{11.2})$$

$$\underline{x}_1 + B \underline{x}_2 | \underline{x}_2 \sim N(\underline{\mu}_1 - \Sigma_{12} \Sigma_{22}^{-1} \underline{x}_2, \Sigma_{11.2})$$

$$\underline{x}_1 | \underline{x}_2 \sim N(\underline{\mu}_1 - B \underline{x}_2 - \Sigma_{12} \Sigma_{22}^{-1} \underline{x}_2, \Sigma_{11.2}) \equiv (\underline{\mu}_1 + \Sigma_{12} \Sigma_{22}^{-1} (\underline{x}_2 - \underline{\mu}_2), \Sigma_{11.2})$$

MULTIPLE & PARTIAL

CORRELATION COEFFICIENTS

In Regression theory,

For ~~multiple~~ simple linear regression model $Y = \beta_0 + \beta_1 X + e$

$$\hat{\beta}_1 = \frac{\text{cov}(X, Y)}{\text{var}(Y)} = \frac{\sigma_{12}}{\sigma_{22}} = \sigma_{12} \sigma_{22}^{-1}$$

distr. of

Last time, we have derived $Y_1 = X_1 + B X_2$ with
 $B = -\Sigma_{12} \Sigma_{22}^{-1}$

Note that, the two terms are comparable.

In the last conditional distr., the conditional mean was obtained as -

$$E(X_1 | X_2) = \mu_1 + B(X_2 - \mu_2) \\ = \mu_1 + \Sigma_{12} \Sigma_{22}^{-1} (X_2 - \mu_2)$$

So, $\Sigma_{12} \Sigma_{22}^{-1}$ can be said the matrix of regression coefficients of X_1 on X_2 . $\therefore \boxed{\beta = \Sigma_{12} \Sigma_{22}^{-1}}$

The $(i, k+1)$ th element of β is ~~$\beta_{ik, k+1, \dots, p}$~~
 $\beta_{ik, k+1, \dots, p}$.

The (i, j) th element of $\Sigma_{11.2}$ is $\sigma_{ij, k+1, \dots, p}$.
 $\downarrow$
 Partial Covariance

Partial Correlation coefficient is -

$$\frac{\sigma_{ij, k+1, \dots, p}}{\sqrt{\sigma_{ii, k+1, \dots, p} \cdot \sigma_{jj, k+1, \dots, p}}}$$

RESULT

16/07/2024

1. Let $\underline{x}_1, \dots, \underline{x}_N$ be N vectors, and $\underline{b}$ be any vector. Then-

$$\sum_{\alpha=1}^N (\underline{x}_\alpha - \underline{b})(\underline{x}_\alpha - \underline{b})' = \sum_{\alpha=1}^N (\underline{x}_\alpha - \bar{\underline{x}})(\underline{x}_\alpha - \bar{\underline{x}})' + N(\bar{\underline{x}} - \underline{b})(\bar{\underline{x}} - \underline{b})'$$

Derivation

$$\begin{aligned} & \sum_{\alpha=1}^N (\underline{x}_\alpha - \underline{b})(\underline{x}_\alpha - \underline{b})' \\ &= \sum_{\alpha=1}^N (\underline{x}_\alpha - \bar{\underline{x}} + \bar{\underline{x}} - \underline{b})(\underline{x}_\alpha - \bar{\underline{x}} + \bar{\underline{x}} - \underline{b})' \\ &= \sum_{\alpha=1}^N (\underline{x}_\alpha - \bar{\underline{x}})(\underline{x}_\alpha - \bar{\underline{x}})' + \sum_{\alpha=1}^N (\bar{\underline{x}} - \underline{b})(\bar{\underline{x}} - \underline{b})' \\ & \quad + 2 \sum_{\alpha=1}^N (\underline{x}_\alpha - \bar{\underline{x}})(\bar{\underline{x}} - \underline{b})' \end{aligned}$$

2. Suppose, in the last result $\underline{b} = \underline{\mu}$. Consider the quantity

$$\begin{aligned} & \sum_{\alpha=1}^N (\underline{x}_\alpha - \underline{\mu})' \Sigma^{-1} (\underline{x}_\alpha - \underline{\mu}) \\ &= \text{tr} \left[\sum_{\alpha=1}^N \Sigma^{-1} (\underline{x}_\alpha - \underline{\mu})(\underline{x}_\alpha - \underline{\mu})' \right] \\ &= \text{tr} \Sigma^{-1} [A + N(\bar{\underline{x}} - \underline{\mu})(\bar{\underline{x}} - \underline{\mu})'] \\ & \quad \text{where } A = \sum_{\alpha=1}^N \frac{(\underline{x}_\alpha - \bar{\underline{x}})(\underline{x}_\alpha - \bar{\underline{x}})'}{(\underline{x}_\alpha - \bar{\underline{x}})'(\underline{x}_\alpha - \bar{\underline{x}})} \\ &= \text{tr} \Sigma^{-1} A + N(\bar{\underline{x}} - \underline{\mu})' \Sigma^{-1} (\bar{\underline{x}} - \underline{\mu}). \end{aligned}$$

LEMMA

3. If D is p.d. of order $p \times p$, the maximum of

$$f(G) = -N \log |G| - \text{tr} G^{-1} D.$$

w.r.t. a pd matrix G exists & occurs at

$$G = \frac{1}{N} D.$$

► CHOLESKY DECOMPOSITION

If A is p.d., $\exists$ a unique lower triangular matrix T ($t_{ij} = 0 \quad +1 < j$) with positive diagonal elements. Such that $A = TT'$.

[e.g.] $T = \begin{pmatrix} a & 0 & 0 \\ b & d & 0 \\ c & e & f \end{pmatrix}$ $|T|^2 = \begin{vmatrix} a^2 & 0 & 0 \\ ab+bd & d^2 & 0 \\ ac+be+cf & de+ef & f^2 \end{vmatrix}$

$$\text{tr}(TT') = a^2 + b^2 + c^2 + d^2 + e^2 + f^2 = a^2 d^2 f^2$$

$$\log |T|^2 = \log a^2 + \log d^2 + \log f^2$$

Proof Let, $D = EE'$ (D is p.d.).

$$\text{and } E' G^{-1} E = H$$

$$\text{and } G = EH^{-1}E'$$

$$\text{So, } |G| = |E| |H^{-1}| |E'|$$

$$= |H^{-1}| |EE'|$$

$$= \frac{|D|}{|H|}$$

$$\text{tr}(G^{-1}D) = \text{tr}(G^{-1}EE')$$

$$= \text{tr}(E'G^{-1}E) = \text{tr}(H)$$

$$f(G) = -N \log |G| - \text{tr} G^{-1}D$$

$$= -N \log |D| + N \log |H| - \text{tr} H$$

$$= -N \log |D| + N \log |TT'| - \text{tr} |TT'| \quad [T = \text{chol}(H)]$$

$$= -N \log |D| + N \log |T|^2 - \text{tr} |T+T'|$$

$$= -N \log |D| + \sum_{i=1}^p N \log t_{ii}^2 + \dots = \sum_{i=1}^p t_{ii}^2$$

$$= -N \log |D| + \sum_{i=1}^p (N \log t_{ii}^2 - t_{ii}^2)$$

$$= -N \log |D| + \sum_{i=1}^p (N \log t_{ii}^2 - t_{ii}^2)$$

$$= -N \log |D| + \sum_{i=1}^p (N \log t_{ii}^2 - t_{ii}^2)$$

$$\frac{\partial f}{\partial t_{ii}^2} = \frac{N}{t_{ii}^2} - 1 = 0 \Rightarrow N = t_{ii}^2$$

$$\frac{\partial^2 f}{\partial (t_{ii}^2)^2} = -\frac{N}{t_{ii}^4} \Rightarrow \frac{\partial^2 f}{\partial (t_{ii}^2)^2} \bigg|_{t_{ii}^2=N} = -\frac{1}{N} < 0$$

Putting, $N = t_{ii}^2$ then $H = N \cdot I_p$

then $G = E H^{-1} E'$

$$= \frac{1}{N} E E'$$

$$= \frac{1}{N} D$$

~~f(G)~~ So, at $G = \frac{1}{N} D$

$$f\left(\frac{1}{N} D\right) = -N \log \left|\frac{D}{N}\right| + N \text{tr}(D^{-1} D)$$

$$= -N \log |D| + p N \log N - N p$$

MLE

$$\underline{x}_1, \underline{x}_2, \dots, \underline{x}_N \sim N_p(\underline{\mu}, \Sigma), \quad N > p$$

$$l = \frac{1}{(2\pi)^{\frac{Np}{2}} |\Sigma|^{\frac{N}{2}}} \exp \left[-\frac{1}{2} \sum_{\alpha=1}^N (\underline{x}_\alpha - \underline{\mu})' \Sigma^{-1} (\underline{x}_\alpha - \underline{\mu}) \right]$$

$$\begin{aligned} \log l &= -\frac{Np}{2} \log 2\pi - \frac{N}{2} \log |\Sigma| - \frac{1}{2} \sum_{\alpha=1}^N (\underline{x}_\alpha - \underline{\mu})' \Sigma^{-1} (\underline{x}_\alpha - \underline{\mu}) \\ &= -\frac{Np}{2} \log 2\pi - \frac{N}{2} \log |\Sigma| - \frac{1}{2} \text{tr} \Sigma^{-1} A \\ &\quad - \frac{N}{2} (\bar{\underline{x}} - \underline{\mu})' \Sigma^{-1} (\bar{\underline{x}} - \underline{\mu}) \end{aligned}$$

(I) (II) (III)

Minimise (III) $\Sigma > 0$, so $\Sigma^{-1} > 0$, $(\bar{\underline{x}} - \underline{\mu})' \Sigma^{-1} (\bar{\underline{x}} - \underline{\mu}) \geq 0$.

Minimizing the term, we get $\hat{\underline{\mu}}_{MLE} = \bar{\underline{x}}$.

Using the lemma, $f(\Sigma) = -N \log |\Sigma| - \text{tr} \Sigma^{-1} A$

Max. occurs at $\Sigma = A/N$.

$$\begin{aligned} &= \frac{1}{N} \sum_{\alpha=1}^N (\underline{x}_\alpha - \bar{\underline{x}}) (\underline{x}_\alpha - \bar{\underline{x}})' \\ &= \hat{\Sigma}_{MLE} \end{aligned}$$

$$E(\hat{\underline{\mu}}) = E(\bar{\underline{x}}) = E \begin{pmatrix} \bar{x}_1 \\ \bar{x}_2 \\ \vdots \\ \bar{x}_p \end{pmatrix} = \begin{pmatrix} \mu_1 \\ \mu_2 \\ \vdots \\ \mu_p \end{pmatrix} = \underline{\mu}$$

Define $S = A/N-1$

$$S = \begin{pmatrix} s_{11} & s_{12} & & \\ & s_{22} & & \\ & & \ddots & \\ & & & s_{pp} \end{pmatrix}$$

$$E(S) = \Sigma$$

MLE is not unbiased, ~~but~~ S is unbiased

17/01/2024

DISTRIBUTION OF SAMPLE MEAN VECTOR

$$\underline{x} \sim N_p(\underline{\mu}, \Sigma).$$

$$f(\underline{x}_1, \dots, \underline{x}_N) = \frac{1}{(2\pi)^{Np/2} |\Sigma|^{N/2}} \exp \left[-\frac{1}{2} \sum_{\alpha=1}^N (\underline{x}_\alpha - \underline{\mu})' \Sigma^{-1} (\underline{x}_\alpha - \underline{\mu}) \right]$$

Apply the transformation

$\underline{x}^* = \underline{x} \underline{Q}$, $\underline{Q}$ is orthogonal matrix
s.t. first column elements
are $1/\sqrt{N}$ and all others
are q_{ij}

$$\underline{Q} = \begin{pmatrix} 1/\sqrt{N} & & \\ & \ddots & \\ & & 1/\sqrt{N} & q_{ij} \end{pmatrix} \quad \underline{x} = \begin{pmatrix} x_{11} & \dots & x_{1N} \\ \vdots & & \vdots \\ x_{p1} & \dots & x_{pN} \end{pmatrix}$$

Now

$$\underline{x}^* = \underline{x} \underline{Q} = \begin{bmatrix} \sqrt{N} \bar{\underline{x}} & \underline{Y} \end{bmatrix}$$

Reduced matrix.

$$\underline{x}^* \underline{x}^{*'} = \underline{x} \underline{Q} \underline{Q}' \underline{x}' = \underline{x} \underline{x}'$$

$$\underline{x}^* \underline{x}^{*'} = N \bar{\underline{x}} \bar{\underline{x}}' + \underline{Y} \underline{Y}'$$

$$\therefore \underline{Y} \underline{Y}' = \underline{x}^* \underline{x}^{*'} - N \bar{\underline{x}} \bar{\underline{x}}'$$

$$= \underline{x} \underline{x}' - N \bar{\underline{x}} \bar{\underline{x}}'$$

$$\sum_{\alpha=1}^N \underline{y}_\alpha \underline{y}_\alpha' = \sum_{\alpha=1}^N (\underline{x}_\alpha - \bar{\underline{x}}) (\underline{x}_\alpha - \bar{\underline{x}})'$$

$$= A = (N-1) S. \quad [\text{See last page notations}]$$

The likelihood is -

$$f(\underline{x}_1, \dots, \underline{x}_N) = \frac{1}{(2\pi)^{Np/2} |\Sigma|^{N/2}} \exp \left[-\frac{1}{2} \sum_{\alpha=1}^N (\underline{x}_\alpha - \underline{\mu})' \Sigma^{-1} (\underline{x}_\alpha - \underline{\mu}) \right]$$

$$= k^* \exp \left[-\frac{1}{2} + \pi \Sigma^{-1} \sum_{\alpha=1}^N (\underline{x}_\alpha - \underline{\mu})' (\underline{x}_\alpha - \underline{\mu}) \right]$$

$$= k^* \exp \left[-\frac{1}{2} \text{tr} \Sigma^{-1} \left\{ \underline{Y} \underline{Y}' + N(\underline{\bar{x}} - \underline{\mu})^E (\underline{\bar{x}} - \underline{\mu})' \right\} \right]$$

$$= \frac{1}{(2\pi)^{Np/2} |\Sigma|^{N/2}} \exp \left[-\frac{1}{2} \text{tr} \Sigma^{-1} \underline{Y} \underline{Y}' \right] \exp \left[-\frac{1}{2} (\underline{\bar{x}} - \underline{\mu})' \left(\frac{\Sigma}{N} \right)^{-1} (\underline{\bar{x}} - \underline{\mu}) \right]$$

$$= \frac{1}{(2\pi)^{Np/2} |\Sigma|^{N/2}} \exp \left[-\frac{1}{2} \underline{Y}' \Sigma^{-1} \underline{Y} \right] \exp \left[-\frac{1}{2} (\underline{\bar{x}} - \underline{\mu})' \left(\frac{\Sigma}{N} \right)^{-1} (\underline{\bar{x}} - \underline{\mu}) \right]$$

$\Rightarrow \underline{\bar{x}}$ & $\underline{Y}$ are independent. — (i)

$$(ii) \underline{\bar{x}} \sim N_p(\underline{\mu}, \frac{\Sigma}{N})$$

$$(iii) \underline{Y}_i \sim N(0, \Sigma), i = 1(1)N$$

~~Also, $\underline{Y}$ is~~

The matrix $\underline{Y}$ follows Wishart distribution.

$$f_{\underline{Y}}(\underline{A}) = \frac{|\underline{A}|^{N-p-1/2} \exp \left[-\frac{1}{2} \text{tr} \Sigma^{-1} \underline{A} \right]}{2^{Np/2} \pi^{p(N-1)/2} |\Sigma|^{N/2} \prod_{i=1}^{N-1} \Gamma \left(\frac{N-i}{2} \right)}$$

Now, if $\underline{x} \sim N_p(\underline{\mu}, \Sigma)$

$$\underline{x}' \Sigma^{-1} \underline{x} \sim \text{Non-central } \chi^2(p, \underline{\mu}' \Sigma^{-1} \underline{\mu})$$

Let, $\underline{c}$ be a n.s. matrix s.t. $\underline{c} \Sigma \underline{c}' = \underline{I}$.

$$\underline{z} = \underline{c} \underline{x} \sim N(\underline{c} \underline{\mu}, \underline{c} \Sigma \underline{c}')$$

$$\sim N(\underline{\lambda}, \underline{I})$$

$$\begin{aligned} \underline{x}' \Sigma^{-1} \underline{x} &= \underline{z}' \underline{c}'^{-1} \Sigma^{-1} \underline{c} \underline{z} = \underline{z}' (\underline{c} \Sigma \underline{c}')^{-1} \underline{z} \\ &= \underline{z}' \underline{z} \end{aligned}$$

Then distribution of $\underline{x}' \Sigma^{-1} \underline{x} = \sum z_i^2 \sim \text{N.C. } \chi^2(p, \sum \lambda_i^2)$

$$\underline{(\bar{x} - \underline{\mu})' \left(\frac{\Sigma}{n} \right)^{-1} (\bar{x} - \underline{\mu}) \sim \chi^2(p)}$$

$$\sqrt{n} (\bar{x} - \underline{\mu}) \sim N(0, \Sigma)$$

$$\boxed{N (\bar{x} - \underline{\mu})' \Sigma^{-1} (\bar{x} - \underline{\mu}) \sim \chi^2(p)}$$

For unknown Σ , to test $H_0: \underline{\mu} = \underline{\mu}_0$, we need Hotelling's T^2 -statistic.

23/07/2024

Theorem $\rightarrow$ If A & B are two real symmetric matrix st. B is pd, then there exists a n.s. matrix P s.t. $P'AP = \text{diag}(\lambda_1, \lambda_2, \dots, \lambda_n)$, $P'BP = I_n$ where $\lambda_1, \dots, \lambda_n$ are the ~~roots~~ of the equation $|A - \lambda B| = 0$.

Proof $\rightarrow$ If B is pd., $\exists$ a n.s. matrix Q s.t. $Q'BQ = I$.
Consider $Q'AQ$, real symmetric.
So by spectral decomposition theorem, $\exists$ an orthogonal matrix H s.t.

$$H'(Q'AQ)H = D = \text{diag}(\lambda_1, \lambda_2, \dots, \lambda_n)$$

where $\lambda_1, \dots, \lambda_n$ are the roots of $|Q'AQ - \lambda I| = 0$.

$$\text{Let, } P = QH \text{ (P is n.s.)}$$

$$\Rightarrow P'AP = D = \text{diag}(\lambda_1, \dots, \lambda_n)$$

[Q is n.s. $\Rightarrow P$ is also n.s.]

$$Q'BQ = I \text{ or, } H'QBQ^{-1} = H'H$$

$$\text{or } P'BP = I \quad (*)$$

Wishart distribution

$$\underline{x}_1, \underline{x}_2, \dots, \underline{x}_n \sim N(\underline{\mu}, \Sigma)$$

$$\hat{\underline{\mu}} = \bar{\underline{x}}, \quad \hat{\Sigma} = S = \frac{A}{n-1} = \frac{A}{n}$$

$$A = \sum_{\alpha=1}^n (\underline{x}_\alpha - \bar{\underline{x}})(\underline{x}_\alpha - \bar{\underline{x}})', \quad n = N-1$$

(P.D. matrix) $(N-1)S = A$

$$S = \begin{pmatrix} s_{11} & s_{12} & \dots & s_{1p} \\ & s_{22} & & \\ & & \ddots & \\ s_{p1} & s_{p2} & \dots & s_{pp} \end{pmatrix} \rightarrow \frac{p(p+1)}{2} \text{ unique elements.}$$

Let, $x_1, \dots, x_N$ ($N > p$) are ind. of each other and they follow $N(\underline{\mu}, \Sigma)$ distribution. If a random pd matrix A , then the PDF -

$$f(A) = \frac{|A|^{n-p-1/2} \exp\left(-\frac{1}{2} \text{tr} \Sigma^{-1} A\right)}{2^{np/2} |\Sigma|^{n/2} \pi^{p(p-1)/4} \prod_{i=0}^{p-1} \Gamma\left(\frac{n-i}{2}\right)}$$

Then A is said to have a Wishart distⁿ with parameter Σ and n .

$$\text{i.e. } \boxed{A \sim W(\Sigma, n)}$$

Theorem: If $\gamma_\alpha \sim N_p(0, \Sigma)$, $\alpha = 1, \dots, n$ and are independently distributed then the density of $A = \sum_{\alpha=1}^n \gamma_\alpha \gamma_\alpha' \sim W(\Sigma, n)$ for pd A & $n > p$. [no density if $n < p$]

$$\begin{aligned} \star \quad & |B'AB - \lambda I| = 0 \\ \Rightarrow & |B'AB - \lambda B'B B| = 0 \\ \Rightarrow & |B'| |A - \lambda B| |B| = 0 \\ \Rightarrow & |B B'| |A - \lambda B| = 0 \\ \Rightarrow & |A - \lambda B| = 0 \end{aligned}$$

Characteristic Function

$$\gamma_\alpha \stackrel{w}{\sim} N(0, \Sigma), \alpha = 1(1)n$$

$$A = \sum_{\alpha=1}^n \gamma_\alpha \gamma_\alpha'$$

$$\begin{aligned} \text{characteristic function!!} \quad \phi(t) &= E[\exp(i \text{tr} A t)] \\ &= E\left[\exp\left(i \text{tr} \sum_{\alpha=1}^n \gamma_\alpha \gamma_\alpha' t\right)\right] \\ &= E\left[\exp\left(i \sum_{\alpha=1}^n \gamma_\alpha' t \gamma_\alpha\right)\right] \end{aligned}$$

$$= \prod_{\alpha=1}^n E [\exp(i \underline{y}'_\alpha \oplus \underline{y}_\alpha)]$$

$$= \left\{ E [\exp(i \underline{y}' \oplus \underline{y})] \right\}^n$$

Consider the quantity,

$$E [\exp(i \underline{y}' \oplus \underline{y})]$$

Consider Σ^{-1} and Θ real, $\exists$ a real

n.s. matrix P s.t. $P' \Sigma^{-1} P = I$ and

$$P' \oplus P = \text{diag}(\lambda_1, \lambda_2, \dots, \lambda_p) = D$$

where $\lambda_1, \dots, \lambda_p$ are the characteristic roots

$$\text{of } |\oplus - \lambda I| = 0$$

$$\text{Now, } \underline{z} \sim N(0, I)$$

Consider a linear transformation, $\underline{y} = P \underline{z}$, z_j 's are independently distributed.

$$E [\exp(i \underline{y}' \oplus \underline{y})]$$

$$= E [\exp(i \underline{z}' P' \oplus P \underline{z})]$$

$$= E [\exp(i \sum_{j=1}^p \lambda_j z_j^2)] \quad [z_j \sim N(0, 1)]$$

$$= \prod_{j=1}^p (1 - 2i\lambda_j)^{-1/2}$$

$$= |I - 2iD|^{-1/2}$$

$$= |P' \Sigma^{-1} P - 2i P' \oplus P|^{-1/2}$$

$$= [P' \Sigma^{-1} P - 2i \oplus P]^{-1/2}$$

$$= [P' \Sigma^{-1} P - 2i \oplus P]^{-1/2}$$

$$= \frac{|I - 2i \oplus \Sigma|^{-1/2}}{|P' P|^{1/2}} = \frac{|I - 2i \oplus \Sigma|^{-1/2}}{|I|^{1/2}}$$

$$= |I - 2i \oplus \Sigma|^{-1/2}$$

$$\phi(\oplus) = |I - 2i \oplus \Sigma|^{-1/2}$$

► Additive Property

If $A_j (j=1, \dots, r)$ i.i.d $W(\Sigma, n_j)$.
 then, $A = \sum_{j=1}^r A_j \sim W(\Sigma, \sum_{j=1}^r n_j)$.

Proof $\phi_{A_j}(\Theta) = |I - 2i\Theta\Sigma|^{-n_j/2}$

$$\begin{aligned}\phi_A(\Theta) &= \phi_{\sum_{j=1}^r A_j}(\Theta) = \prod_{j=1}^r \phi_{A_j}(\Theta) \\ &= |I - 2i\Theta\Sigma|^{-\sum_{j=1}^r n_j/2} \\ &= |I - 2i\Theta\Sigma|^{-n/2} \dots |I - 2i\Theta\Sigma|^{-n_j/2}\end{aligned}$$

Hence, By uniqueness theorem,

$A \sim W(\Sigma, \sum_{j=1}^r n_j)$. [This is also called Reproductive Property]

24/01/2024

► DISTRIBUTION OF LINEAR TRANSFORMATION

If $A \sim W(\Sigma, n)$, $C: p \times p$ n.s. matrix then for
 $B = C'AC \sim W(C'\Sigma C, n)$ where C is s.t. $\underline{Z}_\alpha = C'\underline{Y}_\alpha$.

$A = \Sigma \underline{Y}_\alpha \underline{Y}_\alpha'$
 $\underline{Y}_\alpha \sim N_p(0, \Sigma) \quad \forall \alpha = 1, \dots, n$
 $\underline{Z}_\alpha = C'\underline{Y}_\alpha$

$B = C'AC = C'(\Sigma \underline{Y}_\alpha \underline{Y}_\alpha')C = \Sigma (C'\underline{Y}_\alpha)(C'\underline{Y}_\alpha)' = \Sigma \underline{Z}_\alpha \underline{Z}_\alpha'$

$\underline{Y}_\alpha \sim N(0, \Sigma) \Rightarrow \underline{Z}_\alpha \sim N(0, C'\Sigma C)$

$\therefore \Sigma \underline{Z}_\alpha \underline{Z}_\alpha' \sim W(C'\Sigma C, n)$

► MARGINAL DISTRIBUTION

$$A = \sum \underline{y}_\alpha \underline{y}_\alpha', \quad \underline{y}_\alpha \sim N(0, \Sigma), \quad \underline{y}_\alpha = \begin{pmatrix} y_{\alpha(1)} \\ y_{\alpha(2)} \end{pmatrix}$$

where, $A = \begin{pmatrix} A_{11} & A_{12} \\ A_{21} & A_{22} \end{pmatrix}$

$$\Sigma = \begin{pmatrix} \Sigma_{11} & \Sigma_{12} \\ \Sigma_{21} & \Sigma_{22} \end{pmatrix}$$

So, $A_{11} \sim W(\Sigma_{11}, n)$

~~Let~~ $\begin{cases} \underline{y}_\alpha \sim N(\underline{\mu}, \Sigma) \\ \& A_{11} = \sum_{\alpha=1}^n \underline{y}_\alpha^{(1)} \underline{y}_\alpha^{(1)'} \sim W(\Sigma_{11}, n) \end{cases}$

In general,

$$\underline{y}_\alpha = \begin{bmatrix} y_{\alpha(1)} \\ \vdots \\ y_{\alpha(p)} \end{bmatrix}$$

$$A = \begin{bmatrix} A_{11} & A_{12} & \dots & A_{1p} \\ \vdots & A_{22} & \dots & \vdots \\ A_{p1} & \dots & A_{pp} \end{bmatrix} \quad \Sigma = \begin{bmatrix} \Sigma_{11} & \Sigma_{12} & \dots & \Sigma_{1p} \\ \vdots & \vdots & \ddots & \vdots \\ \Sigma_{p1} & \dots & \dots & \Sigma_{pp} \end{bmatrix}$$

g) $\Sigma_{ij} = 0 \forall i \neq j$ and $A \sim W(\Sigma, n)$ then

$A_{11}, A_{22}, \dots, A_{pp}$ are independently distributed with

$A_{jj} \sim W(\Sigma_{jj}, n)$

Also,

g) $\Sigma_{ij} = 0 \Rightarrow \underline{y}_\alpha^{(1)}, \underline{y}_\alpha^{(2)}, \dots, \underline{y}_\alpha^{(p)} \quad \alpha = 1, \dots, n$

are independently distributed.

$A_{11} = \sum \underline{y}_\alpha^{(1)} \underline{y}_\alpha^{(1)'}, \dots, A_{pp} = \sum \underline{y}_\alpha^{(p)} \underline{y}_\alpha^{(p)'}$

are also independent, $A_{jj} \sim W_{p_j}(\Sigma_{jj}, n) \quad j=1, \dots, p$

Consider the quantity,

$$A_{22.1} = A_{22} - A_{21} A_{11}^{-1} A_{12}$$

$$\underline{y}_\alpha \sim N(0, \Sigma)$$

$$\rightarrow = \begin{pmatrix} y_\alpha^{(1)} \\ y_\alpha^{(2)} \end{pmatrix} \quad A = \sum y_\alpha y_\alpha' = Y^T Y \quad \text{where } Y = \begin{pmatrix} y_1' \\ \vdots \\ y_n' \end{pmatrix} = \begin{pmatrix} y_1 \\ y_2 \end{pmatrix}$$

$$\text{Now, } A_{22.1} = A_{22} - A_{21} A_{11}^{-1} A_{12}$$

$$= y_2' y_2 - y_2' y_1 A_{11}^{-1} y_1' y_2$$

$$= y_2' [I - y_1 A_{11}^{-1} y_1'] y_2$$

$$= y_2' P y_2$$

$$P = I - y_1 A_{11}^{-1} y_1'$$

Symmetric, idempotent

$$\text{rank}(P) = n - q$$

$$y_1 P = 0$$

$$\text{Define, } y_{2.1} = y_2 - y_1 \Sigma_{11}^{-1} \Sigma_{12} \quad [\equiv y_2 | y_1 = y_1]$$

$$\text{Cov}(y_{2.1}, y_1) = (\Sigma_{12} - \Sigma_{11} \Sigma_{11}^{-1} \Sigma_{12}) = 0$$

$$y_{2.1}' P y_{2.1} = y_2' y_2 - y_2' y_1 A_{11}^{-1} y_1' y_2 = y_2' P y_2 = A_{22.1}$$

[Derive]

$$\text{Therefore } y_{2.1} \sim N(\mu_{2.1}, \Sigma_{11.2})$$

$$\& \sum_{\alpha=1}^n y_{2.1} y_{2.1}' \sim W(\Sigma_{11.2}, n)$$

GENERALIZED VARIANCE

Multivariate analogue of σ^2 is $|\Sigma|$. ($|\Sigma|$ also).

Sample version: $|S| = \frac{1}{N-1} \sum_{\alpha=1}^N (\underline{x}_{\alpha} - \bar{\underline{x}})(\underline{x}_{\alpha} - \bar{\underline{x}})'$

Result // Given a random sample $x_1, \dots, x_N$ from $N_p(\mu, \Sigma)$, the distribution of $\frac{|S|}{(N-1)^p}$ times the product of independent factors, where the i th factor follows χ^2_{n-i} distribution.

i.e. $f(|S|) \sim \frac{|S|}{N^p} f(\chi^2_1) \dots f(\chi^2_{N-p+1})$

ASYMPTOTIC DISTRIBUTION

$$\sqrt{N} \left(\frac{|S|}{|\Sigma|} - 1 \right) \sim N(0, 2p)$$

$$E(|S|) = \frac{|\Sigma|}{N^p} \prod_{i=1}^p E(\chi^2_{N-i})$$

$$= \frac{|\Sigma|}{N^p} \prod_{i=1}^p (N-i)$$

COCHRAN'S THEOREM

Suppose $\underline{y}_1, \dots, \underline{y}_N \stackrel{iid}{\sim} N(0, \Sigma)$

Rank $(C_i) = m$, suppose $C_i = ((C_{\alpha\beta}^i))$

$$\sum_{i=1}^m C_i = I_N$$

$$B_i = \sum_{\alpha=1}^N \sum_{\beta=1}^N C_{\alpha\beta}^i \underline{y}_{\alpha} \underline{y}_{\beta}'$$

$$\sum_{i=1}^m B_i = \sum_{\alpha=1}^N \underline{y}_{\alpha} \underline{y}_{\alpha}'$$

A necessary and sufficient condition that $B_1, \dots, B_m$ are independently distributed with $B_i \sim W(\Sigma, n_i)$.

$$\sum_{i=1}^m n_i = N$$

INVERTED WISHART DISTRIBUTION

$$A \sim W(\Sigma, n)$$

$$B = A^{-1} \sim f(B) = \frac{|\Sigma^{-1}|^{n/2} |B|^{-\frac{n+p-1}{2}}}{2^{np/2} |\Sigma|^{(n-p)/2} \pi^{p(p-1)/4} \prod_{i=1}^p \Gamma\left(\frac{n-i}{2}\right)}$$

[Derive]

$$\exp\left[-\frac{1}{2} + \pi \Sigma^{-1} B^{-1}\right]$$

► $y_\alpha \sim N(0, \Sigma)$

$$\Sigma y_\alpha y_\alpha' \sim W(\Sigma, n)$$

$x_\alpha \sim N(\mu, \Sigma)$

$$A = \Sigma x_\alpha x_\alpha' \sim \text{Non-central Wishart}(\Sigma, n, \Sigma^{-1} \mu \mu')$$

► $A \sim W(\Sigma, n)$

$$E(A) = n\Sigma \quad \& \quad E(A^{-1}) = \frac{\Sigma^{-1}}{n-p-1}$$

GENERALISED T^2 STATISTIC

29/01/2024.

Univariate $H_0: \mu = \mu_0, x_1, \dots, x_n \sim N(\mu, \sigma^2)$

$$t = \frac{\bar{x} - \mu_0}{s/\sqrt{n}} \sim t_{n-1}$$

Multivariate $H_0: \underline{\mu} = \underline{\mu}_0$

$$\underline{x}_1, \dots, \underline{x}_n \sim N_p(\underline{\mu}, \underline{\Sigma})$$

$$T^2 = N (\bar{\underline{x}} - \underline{\mu})' \underline{\Sigma}^{-1} (\bar{\underline{x}} - \underline{\mu})$$

~~Hotelling~~ T^2 statistic

Sample version: (6)

Result For any $|B| \neq 0$.

$$\begin{vmatrix} B & C \\ D & E \end{vmatrix} = \begin{vmatrix} B & C \\ D - EB^{-1}C & E - CB^{-1}C \end{vmatrix}$$

$$= \begin{vmatrix} B & 0 \\ D & E - DB^{-1}C \end{vmatrix}$$

$$= |B| |E - DB^{-1}C|$$

e.g.

$$\begin{vmatrix} A & \sqrt{N}(\bar{\underline{x}} - \underline{\mu}) \\ \sqrt{N}(\bar{\underline{x}} - \underline{\mu})' & I \end{vmatrix}$$

$$= \begin{vmatrix} A & \sqrt{N}(\bar{\underline{x}} - \underline{\mu}) \\ -\sqrt{N}(\bar{\underline{x}} - \underline{\mu})' & I \end{vmatrix}$$

$$= \begin{vmatrix} A & \sqrt{N}(\bar{\underline{x}} - \underline{\mu}) \\ -\sqrt{N}(\bar{\underline{x}} - \underline{\mu})' & I \end{vmatrix} \begin{vmatrix} I & -\sqrt{N}A^{-1}(\bar{\underline{x}} - \underline{\mu})' \\ 0 & I \end{vmatrix}$$

$$= \begin{vmatrix} A & 0 \\ -\sqrt{N}(\bar{\underline{x}} - \underline{\mu})' & I + N(\bar{\underline{x}} - \underline{\mu})A(\bar{\underline{x}} - \underline{\mu})' \end{vmatrix}$$

LIKELIHOOD RATIO TEST

$$\underline{X} \sim N_p(\underline{\mu}, \underline{\Sigma})$$

$\underline{x}_1, \dots, \underline{x}_N \rightarrow \text{sample.}$

$$L = L(\underline{\mu}, \underline{\Sigma}) = \frac{1}{(2\pi)^{Np/2} |\underline{\Sigma}|^{N/2}} \exp \left[-\frac{1}{2} \sum_{\alpha=1}^N (\underline{x}_{\alpha} - \underline{\mu})' \underline{\Sigma}^{-1} (\underline{x}_{\alpha} - \underline{\mu}) \right]$$

The LR test statistic is -

$$\lambda = \frac{\text{Max}_{H_0} L}{\text{Max}_{\Omega} L}$$

$$\hat{\Omega}_{H_0} = (\hat{\underline{\mu}}, \hat{\underline{\Sigma}}) = (\underline{\mu}_0, \hat{\underline{\Sigma}}_w)$$

$$\hat{\Omega}_{\Omega} = (\hat{\underline{\mu}}_{\Omega}, \hat{\underline{\Sigma}}_{\Omega}) = (\bar{\underline{x}}, \frac{\underline{A}}{N})$$

$$\text{where } \begin{cases} \hat{\underline{\Sigma}}_{\Omega} = \frac{1}{N} \sum_{\alpha=1}^N (\bar{\underline{x}}_{\alpha} - \bar{\underline{x}}) (\bar{\underline{x}}_{\alpha} - \bar{\underline{x}})' \\ \hat{\underline{\Sigma}}_w = \frac{1}{N} \sum_{\alpha=1}^N (\underline{x}_{\alpha} - \underline{\mu}_0) (\underline{x}_{\alpha} - \underline{\mu}_0)' \end{cases}$$

$$\text{Max}_{H_0} L = \frac{1}{(2\pi)^{Np/2} |\hat{\underline{\Sigma}}_w|^{N/2}} \exp \left[-\frac{1}{2} \sum_{\alpha=1}^N (\underline{x}_{\alpha} - \underline{\mu}_0)' \hat{\underline{\Sigma}}_w^{-1} (\underline{x}_{\alpha} - \underline{\mu}_0) \right]$$

$$= \frac{1}{(2\pi)^{Np/2} |\hat{\underline{\Sigma}}_w|^{N/2}} \exp \left[-\frac{1}{2} \text{tr} \hat{\underline{\Sigma}}_w^{-1} \sum_{\alpha=1}^N \left\{ (\underline{x}_{\alpha} - \underline{\mu}_0) (\underline{x}_{\alpha} - \underline{\mu}_0)' \right\} \right]$$

$$= \frac{1}{(2\pi)^{\frac{Np}{2}} |\hat{\underline{\Sigma}}_w|^{N/2}} \exp \left[-\frac{N}{2} \text{tr} \hat{\underline{\Sigma}}_w^{-1} \underline{\Sigma}_w \right]$$

$$= \frac{1}{(2\pi)^{Np/2} |\hat{\underline{\Sigma}}_w|^{N/2}} \exp \left[-\frac{Np}{2} \right]$$

We know, $A = \sum_{\alpha=1}^N (\underline{x}_{\alpha} - \bar{x})(\underline{x}_{\alpha} - \bar{x})'$

$$A_0 = \sum_{\alpha=1}^N (\underline{x}_{\alpha} - \underline{\mu}_0)(\underline{x}_{\alpha} - \underline{\mu}_0)'$$

$$= A + N(\bar{x} - \underline{\mu}_0)(\bar{x} - \underline{\mu}_0)' \quad \text{--- (**)}$$

Now,

$$\text{Max}_{\Sigma} L = \frac{1}{(2\pi)^{Np/2} |\hat{\Sigma}_2|^{N/2}} \exp \left[-\frac{1}{2} \sum_{\alpha=1}^N (\underline{x}_{\alpha} - \bar{x})' \hat{\Sigma}_2^{-1} (\underline{x}_{\alpha} - \bar{x}) \right]$$

$$= \frac{1}{(2\pi)^{Np/2} |\hat{\Sigma}_2|^{N/2}} \exp \left[-\frac{1}{2} \text{tr} \left(\hat{\Sigma}_2^{-1} \sum_{\alpha=1}^N \{ (\underline{x}_{\alpha} - \bar{x})(\underline{x}_{\alpha} - \bar{x})' \} \right) \right]$$

$$= \frac{1}{(2\pi)^{Np/2} |\hat{\Sigma}_2|^{N/2}} \exp \left[-\frac{N}{2} \text{tr} \left(\hat{\Sigma}_2^{-1} \hat{\Sigma}_2 \right) \right]$$

$$= \frac{1}{(2\pi)^{Np/2} |\hat{\Sigma}_2|^{N/2}} \exp(-Np/2)$$

$$\therefore \lambda = \frac{|\hat{\Sigma}_2|^{N/2}}{|\hat{\Sigma}_w|^{N/2}} \quad \text{or, } \lambda^{2/N} = \frac{|\hat{\Sigma}_2|}{|\hat{\Sigma}_w|}$$

$$= \frac{\left| \sum_{\alpha=1}^N (\underline{x}_{\alpha} - \bar{x})(\underline{x}_{\alpha} - \bar{x})' \right|}{\left| \sum_{\alpha=1}^N (\underline{x}_{\alpha} - \underline{\mu}_0)(\underline{x}_{\alpha} - \underline{\mu}_0)' \right|}$$

$$= \frac{|A|}{|A|}$$

$$|A + N(\bar{x} - \underline{\mu}_0)(\bar{x} - \underline{\mu}_0)'|$$

$$= \frac{|A|}{|A|} \quad \text{[from **]}$$

$$= \frac{|A| |I + N(\bar{x} - \underline{\mu}_0)(\bar{x} - \underline{\mu}_0)'|}{|A|}$$

$$= \frac{|A| [1 + N(\bar{x} - \underline{\mu}_0)' A^{-1} (\bar{x} - \underline{\mu}_0)]}{|A|}$$

$$\lambda = \frac{2/N}{1 + \frac{T^2}{N-1}}$$

Reject H_0 , if $\lambda \leq \lambda_0$.

$$T^2 = N(\bar{X} - \mu_0)' S^{-1} (\bar{X} - \mu_0)$$

$$= N(N-1) (\bar{X} - \mu_0)' A^{-1} (\bar{X} - \mu_0)$$

λ_0 is chosen s.t. $P(\lambda \leq \lambda_0) = \alpha$ when H_0 is true.

$$\lambda \leq \lambda_0 \Rightarrow \frac{N}{1 + \frac{T^2}{N-1}} \leq \lambda_0$$

$$\Rightarrow \lambda^{-2/N} - 1 \geq \lambda_0^{-2/N} - 1$$

$$\Rightarrow (N-1) (\lambda^{-2/N} - 1) \geq (N-1) (\lambda_0^{-2/N} - 1)$$

$$\Rightarrow (N-1) \left(1 + \frac{T^2}{N-1} - 1\right) \geq (N-1) \left(1 + \frac{T_0^2}{N-1} - 1\right)$$

$$\Rightarrow T^2 \geq T_0^2(\alpha), \text{ where } T_0^2(\alpha) \text{ is a const. to be determined with size condition.}$$

$$\text{Now, } T_0^2 = (N-1) (\lambda_0^{-2/N} - 1)$$

Consider Σ known then $(\bar{X} - \mu_0)' S^{-1} (\bar{X} - \mu_0) \sim \chi_p^2$.

DISTRIBUTION OF T^2

Let, $\underline{Y} \sim N_p(\underline{\mu}, \Sigma)$ and $A = nS \sim W_p(n, \Sigma)$.

$\underline{Y}$ & S are independent.

$$\text{If } T^2 = \underline{Y}' S^{-1} \underline{Y}, \text{ then } \frac{T^2}{n} \sim \frac{n-p+1}{p} \sim F(p, n-p+1, \delta)$$

non-centrality parameter.

$$\text{Proof} \Rightarrow \begin{bmatrix} (a \ b) (a \ b) \\ (c \ d) (c \ d) \end{bmatrix} = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$$

$$\Rightarrow \begin{pmatrix} a^2+b^2 & ac+bd \\ ac+bd & c^2+d^2 \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$$

$$\Rightarrow \begin{cases} a^2+b^2 = 1 = c^2+d^2 \\ ac+bd = 0 \end{cases}$$

2nd original setup

$$\begin{cases} T^2 = N(\underline{\bar{X}} - \underline{\mu_0})' S^{-1} (\underline{\bar{X}} - \underline{\mu_0}) \\ Y = \sqrt{N}(\underline{\bar{X}} - \underline{\mu_0}) \end{cases}$$

Let, D be a non-singular matrix st. $D \Sigma D' = I$.

Define, $\underline{Y}^* = D\underline{Y}$, $S^* = DSD'$, $\underline{Y}^* = D\underline{Y}$, $\underline{Z}^* = D\underline{Z}$.

[Lemma] For any $p \times p$ n.s. matrix C and H and any vector $\underline{u}$, $\underline{u}' H^{-1} \underline{u} = (C\underline{u})' (CHC')^{-1} (C\underline{u})$

$$T^2 = \underline{Y}^{*'} S^{*-1} \underline{Y}^*$$

$$\begin{aligned} &= (D\underline{Y})' (DSD')^{-1} (D\underline{Y}) \\ &= \underline{Y}' D D^{-1} S D^{-1} D \underline{Y} \\ &= \underline{Y}' S \underline{Y} \end{aligned}$$

Now, $\underline{Y}^* \sim N_p(\underline{Y}^*, D \Sigma D' = I)$

and $\underline{Y}^*$ and $n S^*$ are independently distributed as -

$$\sum_{i=1}^n \underline{Z}_i^* \underline{Z}_i^{*'} = \sum_{i=1}^n (D \underline{Z}_i) (D \underline{Z}_i)'$$

each $\underline{Z}_i^* \sim N(0, I)$.

Define an orthogonal matrix $Q_{p \times p}$ such that

first row is $q_{1i} = \frac{Y_i^*}{\sqrt{Y^{*'} Y^*}}$, $i=1, \dots, p$.

and remaining rows can be chosen arbitrarily.

[Lemma: $A_{n \times m}$ ($n > m$) s.t. $A'A = I_m$ then $\exists$ a matrix B , s.t. $(A \ B)$ is orthogonal]

Since, Q depends on y^* (random), it is a random matrix.

Let, $U = QY^*$, $B = Q(n \times p)Q'$.

$U = (U_i)$ $U_1 = \sum_i q_{1i} y_i^* = \sqrt{y^{*'} y^*}$

$U_j = \sum_i q_{ji} y_i^* = \sqrt{y^{*'} y^*} \sum_i q_{ji} q_{1i} = 0$.

$\left[\frac{T^2}{n} = y^* (Q'Q)^{-1} S^{-1} (Q'Q) y^* \right]$
 $= y^* S^{-1} y^* \cdot$

$\frac{T^2}{n} = U' B^{-1} U = (U_1, 0, \dots, 0) \begin{pmatrix} b^{11} & b^{12} & \dots & b^{1p} \\ \vdots & \vdots & \ddots & \vdots \\ b^{p1} & b^{p2} & \dots & b^{pp} \end{pmatrix} \begin{pmatrix} U_1 \\ 0 \\ \vdots \\ 0 \end{pmatrix}$
 $= U_1^2 \cdot b^{11}$

$[B^{-1} = (b^{ij})]$

$B = \begin{bmatrix} b_{11} & b_{12} \\ b_{21} & B_{22} \end{bmatrix} \Rightarrow \frac{1}{b^{11}} = b_{11} - b_{12}' B_{22}^{-1} b_{12}$
 $= b_{11.23 \dots p}$

[Recall: $A = \begin{pmatrix} A_{11} & A_{12} \\ A_{12}' & A_{22} \end{pmatrix}$.

Using that, $A_{11.2} = A_{11} - A_{12} A_{22}^{-1} A_{12}' \sim W_p(n-p+1, \Sigma_{11.2})$

$b_{11.23 \dots p} \sim \chi^2_{(n-p+1)}$

$y^{*'} y^* \sim \text{non-central } \chi^2$

$\frac{T^2}{n} = \frac{U_1^2}{b^{11}} = \frac{y^{*'} y^*}{b_{11.23 \dots p}} \sim \frac{\text{non-central } \chi^2_p}{\chi^2_{(n-p+1)}}$

$\frac{\frac{n-p+1}{p} \cdot \frac{T^2}{n}}{\frac{N-p}{p} \cdot \frac{T^2}{N-1}} \sim \text{non-central } F(p, n-p+1, \gamma' \Sigma^{-1} \gamma)$
 $\sim \text{n.c.F. } (p, N-p, \gamma' \Sigma^{-1} \gamma)$

T^2 distribution with $n (= N-1)$ d.f.

of $H_0: \underline{\mu} = \underline{\mu}_0$, $\underline{Y} = \sqrt{N}(\bar{\underline{X}} - \underline{\mu}_0)$.

$E(\underline{Y}) = \underline{0}$

① One-sample mean Problem

$X_1, \dots, X_N \stackrel{i.i.d}{\sim} N(\underline{\mu}, \Sigma)$, $\lambda = \frac{1}{1 + \frac{T^2}{N-1}}$

W: $\lambda \leq \lambda_0 \rightarrow T^2 \geq T_0^2(\alpha)$

$T_0^2 = (N-1) (\lambda_0^{-2/N} - 1)$

So, $T_0^2 = \frac{(N-1)p}{N-p} F_{p, N-p}(\alpha)$

$= T_{p, N-1}^2(\alpha)$

$= T_{p, n}^2(\alpha)$

CONFIDENCE REGION

$\underline{\mu}$ s.t., $N(\bar{\underline{X}} - \underline{\mu})' S^{-1} (\bar{\underline{X}} - \underline{\mu}) \leq T_{p, N-1}^2(\alpha)$

Lemma If $B_{p \times p}$ is a n.s. matrix &
 $\underline{v}$: $p \times 1$ vector, then $\underline{v}' B^{-1} \underline{v}$ is
 the non-zero root of $|\underline{v} \underline{v}' - \lambda B| = 0$

Proof Let, λ_1 is a non-zero root of $|\underline{v} \underline{v}' - \lambda B| = 0$
 with characteristic vector $\underline{\beta}$ satisfying $\underline{v} \underline{v}' \underline{\beta} = \lambda \underline{\beta}$
 Since $\lambda \neq 0$, $\underline{v}' \underline{\beta} \neq 0$, we have $\underline{v}' B^{-1} \cdot \underline{v} \underline{v}' \underline{\beta} = \underline{v}' B^{-1} \cdot \lambda \underline{\beta}$

or, $(\underline{v}' B^{-1} \underline{v}) \cdot (\underline{v}' \underline{\beta}) = \lambda_1 (\underline{v}' \underline{\beta})$

Consider, $\underline{V} = \sqrt{N} (\bar{\underline{x}} - \underline{\mu}_0)$, $B = A$.

T^2 statistics $\rightarrow$ computed from $\bar{\underline{x}}$ & A .

$A^{-1} (\bar{\underline{x}} - \underline{\mu}_0) = \underline{b}$ is a solution of $A \underline{b} = \bar{\underline{x}} - \underline{\mu}_0$.

$$= N (\bar{\underline{x}} - \underline{\mu}_0) \underline{b}$$

$\frac{T^2}{N-1}$ is the non-zero root of $|\underline{N} (\bar{\underline{x}} - \underline{\mu}_0) (\bar{\underline{x}} - \underline{\mu}_0)' - \lambda A| = 0$.

① TWO-SAMPLE TEST

$$\begin{aligned} & \underline{x}_1^{(1)}, \underline{x}_2^{(1)}, \dots, \underline{x}_{N_1}^{(1)} \sim N(\underline{\mu}^{(1)}, \Sigma) \\ & \underline{x}_1^{(2)}, \underline{x}_2^{(2)}, \dots, \underline{x}_{N_2}^{(2)} \sim N(\underline{\mu}^{(2)}, \Sigma) \\ & \bar{\underline{x}}^{(1)}, \bar{\underline{x}}^{(2)} \\ & S_1, S_2 \end{aligned}$$

$$H_0: \underline{\mu}^{(1)} = \underline{\mu}^{(2)}$$

Pooled Covariance matrix

$$S = \frac{(N_1 - 1)S_1 + (N_2 - 1)S_2}{N_1 + N_2 - 2}$$

$$= \frac{n_1 S_1 + n_2 S_2}{n_1 + n_2}$$

$$S_1 = \frac{1}{N_1 - 1} \sum_{k=1}^{N_1} (\underline{x}_k^{(1)} - \bar{\underline{x}}^{(1)}) (\underline{x}_k^{(1)} - \bar{\underline{x}}^{(1)})'$$

$$S_2 = \frac{1}{N_2 - 1} \sum_{k=1}^{N_2} (\underline{x}_k^{(2)} - \bar{\underline{x}}^{(2)}) (\underline{x}_k^{(2)} - \bar{\underline{x}}^{(2)})'$$

$$V(\bar{\underline{x}}^{(1)} - \bar{\underline{x}}^{(2)}) = \frac{\Sigma}{N_1} + \frac{\Sigma}{N_2} = \frac{N_1 + N_2}{N_1 N_2} \Sigma$$

$$\Rightarrow V\left(\sqrt{\frac{N_1 N_2}{N_1 + N_2}} (\bar{\underline{x}}^{(1)} - \bar{\underline{x}}^{(2)})\right) = \Sigma$$

Therefore, under H_0 ,

$$\sqrt{\frac{N_1 N_2}{N_1 + N_2}} (\bar{\underline{x}}^{(1)} - \bar{\underline{x}}^{(2)}) \sim N(\underline{0}, \Sigma)$$

$$(N_1 + N_2 - 2) S \sim \sum_{\alpha=1}^{N_1 + N_2 - 2} \bar{z}_\alpha \bar{z}_\alpha' \text{ where } \bar{z}_\alpha \sim N(0, \Sigma)$$

$$T^2 = \frac{N_1 N_2}{N_1 + N_2} (\bar{x}^{(1)} - \bar{x}^{(2)})' S^{-1} (\bar{x}^{(1)} - \bar{x}^{(2)})$$

$$\sim T_{N_1 + N_2 - 2}^2 \text{ (under } H_0 \text{)}.$$

So, under H_0 , $\frac{N_1 + N_2 - p - 1}{p} \cdot \frac{T^2}{N_1 + N_2 - 2} \sim F(p, N_1 + N_2 - p - 1)$
(α).

Confidence Region

Set of vectors 'm' s.t.

$$(\bar{x}^{(1)} - \bar{x}^{(2)} - m)' S^{-1} (\bar{x}^{(1)} - \bar{x}^{(2)} - m) \leq \frac{N_1 + N_2}{N_1 N_2} T_{p, N_1 + N_2 - 2}^2(\alpha).$$

③ Problem of symmetry
(Symmetry property of T^2)

$$H_0: \mu_1 = \mu_2 = \dots = \mu_p.$$

$$\text{alternatively, } H_0: \mu_1 - \mu_2 = 0,$$

$$\mu_2 - \mu_3 = 0,$$

$$\vdots$$

$$\mu_{p-1} - \mu_p = 0.$$

(p-1) linearly independent contrasts.

$$\text{Or, } H_0: C\mu = 0$$

where: C is a $(p-1) \times p$ matrix

$$\text{ \& } \underline{\mu} = (\mu_1, \dots, \mu_p)'$$

$$\text{Now, } \underline{y}_\alpha = C \underline{x}_\alpha \sim N(C\mu, C\Sigma C')$$

if under $H_0: \mu = 0$

$$T_x^2 = N(\bar{x} - 0)' S^{-1} (\bar{x} - 0) = N\bar{x}' S^{-1} \bar{x}$$

$$T_y^2 = N(C\bar{x})' (C S C')^{-1} (C\bar{x}) = N\bar{x}' C' C^{-1} S C^{-1} C \bar{x} = N\bar{x}' S^{-1} \bar{x}.$$

So, basically, T^2 is invariant of a non-singular linear transformation.

Also, T^2 is independent of c .

e.g. >

$$H_0: \mu_1 - \mu_2 + \mu_3 = \mu_4.$$

$$\equiv (1 \ -1 \ 1 \ -1) \begin{pmatrix} \mu_1 \\ \mu_2 \\ \mu_3 \\ \mu_4 \end{pmatrix} = 0$$

$$H_0: C\mu = 0.$$

1) MANHATTAN DISTANCE

$$\Delta^2 = (\underline{\mu}_1 - \underline{\mu}_2)' \Sigma^{-1} (\underline{\mu}_1 - \underline{\mu}_2).$$

Distance b/w two population $N(\underline{\mu}_1, \Sigma), N(\underline{\mu}_2, \Sigma)$

$$\underline{x} = (x_1, \dots, x_p)'$$

$$\underline{y} = (y_1, \dots, y_p)' \in \mathbb{R}^p.$$

$$d_s(\underline{x}, \underline{y}) = \sqrt{(\underline{x} - \underline{y})' \Sigma^{-1} (\underline{x} - \underline{y})}.$$

Properties.

$$(i) \ d_s(\underline{x}, \underline{y}) = d_s(\underline{y}, \underline{x}).$$

$$(ii) \ d_s(\underline{x}, \underline{y}) > 0 \text{ if } \underline{x} \neq \underline{y}$$

$$(iii) \ d_s(\underline{x}, \underline{y}) = 0 \text{ if } \underline{x} = \underline{y}$$

$$(iv) \ d_s(\underline{x}, \underline{y}) \leq d_s(\underline{x}, \underline{z}) + d_s(\underline{z}, \underline{y}).$$

↳ [Triangle Inequality]

RAO'S U STATISTIC

$$\underline{Y} \sim N(\underline{\mu}, \Sigma), A \sim W(n, \Sigma)$$

independent.

Partition

$$\underline{Y} = \begin{pmatrix} \underline{y}_1^{r \times 1} \\ \underline{y}_2^{p-r \times 1} \end{pmatrix}$$

$$\underline{\mu} = \begin{pmatrix} \underline{\mu}_1 \\ \underline{\mu}_2 \end{pmatrix}$$

$$\Sigma = \begin{pmatrix} \Sigma_{11} & \Sigma_{12} \\ \Sigma_{21} & \Sigma_{22} \end{pmatrix}$$

$$A = \begin{pmatrix} A_{11} & A_{12} \\ A_{21} & A_{22} \end{pmatrix}$$

Consider $\Delta_p^2 = \underline{Y}' \Sigma^{-1} \underline{Y}$

$$= (\underline{\mu}_1' \quad \underline{\mu}_2') \begin{pmatrix} \Sigma_{11}^{-1} + \beta' \Sigma_{22}^{-1} \beta & -\beta \Sigma_{22}^{-1} \\ -\Sigma_{22}^{-1} \beta & \Sigma_{22}^{-1} \end{pmatrix} \begin{pmatrix} \underline{\mu}_1 \\ \underline{\mu}_2 \end{pmatrix}$$

$$= \underline{\mu}_1' \Sigma_{11}^{-1} \underline{\mu}_1 + (\underline{\mu}_2 - \beta \underline{\mu}_1)' \Sigma_{22}^{-1} (\underline{\mu}_2 - \beta \underline{\mu}_1)$$

$$\Delta_p^2 = \Delta_q^2 + \dots$$

$$(C, C) \cdot ab = (C, C) \cdot ab \quad (i)$$

$$C + x \cdot i \cdot C = (C, C) \cdot ab \quad (ii)$$

$$C = x \cdot i \cdot C = (C, C) \cdot ab \quad (iii)$$

$$(C, C) \cdot ab + (C, C) \cdot ab = (C, C) \cdot ab \quad (iv)$$

separating

RAO'S U STATISTIC

$$\underline{Y} \sim N(\underline{\mu}, \Sigma), A \sim W(n, \Sigma).$$

independent.

$$\Sigma = \begin{pmatrix} \Sigma_{11} & \Sigma_{12} \\ \Sigma_{21} & \Sigma_{22} \end{pmatrix}$$

Partition

$$\underline{Y} = \begin{pmatrix} \underline{y}_1 \\ \underline{y}_2 \end{pmatrix}$$

$$\underline{\mu} = \begin{pmatrix} \underline{\mu}_1 \\ \underline{\mu}_2 \end{pmatrix}$$

$$\Sigma = \begin{pmatrix} \Sigma_{11} & \Sigma_{12} \\ \Sigma_{21} & \Sigma_{22} \end{pmatrix}$$

$$A = \begin{pmatrix} A_{11} & A_{12} \\ A_{21} & A_{22} \end{pmatrix}$$

Consider $\Delta_p^2 = \underline{y}' \Sigma^{-1} \underline{y}$

$$= (\underline{\mu}_1', \underline{\mu}_2') \begin{pmatrix} \Sigma_{11}^{-1} + \beta' \Sigma_{22}^{-1} \beta & -\beta \Sigma_{22}^{-1} \\ -\Sigma_{22}^{-1} \beta & \Sigma_{22}^{-1} \end{pmatrix} \begin{pmatrix} \underline{\mu}_1 \\ \underline{\mu}_2 \end{pmatrix}$$

$\beta = \Sigma_{21} \Sigma_{11}^{-1}$ (?)

$$= \underline{\mu}_1' \Sigma_{11}^{-1} \underline{\mu}_1 + (\underline{\mu}_2 - \beta \underline{\mu}_1)' \Sigma_{22.1}^{-1} (\underline{\mu}_2 - \beta \underline{\mu}_1)$$

$$\Delta_p^2 = \Delta_q^2 + (\underline{\mu}_2 - \beta \underline{\mu}_1)' \Sigma_{22.1}^{-1} (\underline{\mu}_2 - \beta \underline{\mu}_1)$$

~~7/3/2024~~

05/02/2024

$$T_p^2 = \underline{y}' A^{-1} \underline{y} = \begin{pmatrix} \underline{y}_1 \\ \underline{y}_2 \end{pmatrix}' \begin{pmatrix} A_{11} & A_{12} \\ A_{21} & A_{22} \end{pmatrix} \begin{pmatrix} \underline{y}_1 \\ \underline{y}_2 \end{pmatrix}$$

$$= T_q^2 + (\underline{y}_2 - B \underline{y}_1)' A_{22.1}^{-1} (\underline{y}_2 - B \underline{y}_1)$$

$$[B = A_{21} A_{11}^{-1}]$$

$$\Rightarrow \frac{T_p^2}{n} = \frac{T_q^2}{n} + \underline{z}' \Lambda_{22 \cdot 1}^{-1} \underline{z} \quad [n = N-1]$$

Define,
$$U = \frac{1 + T_q^2/n}{1 + T_p^2/n} \rightarrow \text{Rao's } U\text{-statistic.}$$

and,
$$\frac{n-p+1}{p-q} \cdot \left(\frac{1}{U} - 1 \right) \sim F(p-q, n-p+1) \text{ when } \Delta_p^2 = \Delta_q^2$$

$$\left\{ \begin{array}{l} \Delta_p^2 = \text{Distance b/w 2 populations with all } p \text{ variables} \\ \Delta_q^2 = \text{Distance b/w 2 populations with all } q \text{ variables.} \end{array} \right. \quad [p > q]$$

The null-hypothesis: $H_0: \Delta_p^2 = \Delta_q^2$

First q factors are equally informative.

Last $(p-q)$ factors are not useful (redundant).

Now, $\Delta_p^2 = \Delta_q^2$ if $\underline{\mu}_2 - \beta \underline{\mu}_1 = 0$ or $\underline{\Sigma}_{22 \cdot 1}^{-1} = 0$.
this is not possible

Hence, $\Delta_p^2 = \Delta_q^2$ if $\underline{\mu}_2 = \beta \underline{\mu}_1$

$\rightarrow$ If $\underline{\mu}_1 = 0$, then $\underline{\mu}_2 = 0$.

So, the null hypothesis can be reformulated as -

$H_0: \underline{\mu}_1 = 0 \text{ (if } \underline{\mu}_2 = 0 \text{)}$

$\rightarrow$ The linear combination of $\underline{a}' \underline{x} \quad [\underline{x} = \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}]$
Such that $\underline{x}$ & $\underline{x}_1$ are uncorrelated.

Hence, $\text{cov}(\underline{a}' \underline{x}, \underline{x}_1) = 0$

Or, $\text{cov}(\underline{a}_1' \underline{x}_1 + \underline{a}_2' \underline{x}_2, \underline{x}_1) = 0$

$= \underline{a}_1' \underline{\Sigma}_{11} + \underline{a}_2' \underline{\Sigma}_{21} = 0 \Rightarrow \underline{a}_1' = -\underline{a}_2' \beta$

$$E(\underline{a}' \underline{x}) = \underline{a}' \underline{\mu}_1 + \underline{a}' \underline{\mu}_2 \\ = \underline{a}' (\underline{\mu}_2 - \beta \underline{\mu}_1)$$

$$\text{If } E(\underline{a}' \underline{x}) = 0 \quad \forall \underline{a}_1$$

$$\Rightarrow \underline{\mu}_2 - \beta \underline{\mu}_1 = 0$$

$$\Rightarrow \underline{\mu}_2 = \beta \underline{\mu}_1$$

06/02/2024

MULTIVARIATE BEHREN - FISHER PROBLEM

$$H_0: \underline{\mu}_1 = \underline{\mu}_2$$

We have two independent normal population - one from $N(\underline{\mu}_1, \Sigma_1)$, $N(\underline{\mu}_2, \Sigma_2)$

Σ_1, Σ_2 : unknown & unequal.

Sample: $\underline{x}_\alpha^{(1)}, \alpha = 1(1)N_1$

$\underline{x}_\alpha^{(2)}, \alpha = 1(1)N_2$

So,

$$\underline{\bar{x}}_1 - \underline{\bar{x}}_2 \sim N(\underline{\mu}_1 - \underline{\mu}_2, \frac{\Sigma_1}{N_1} + \frac{\Sigma_2}{N_2})$$

Sample
Covariance
matrix

$$\sum_{\alpha=1}^{N_1} (\underline{x}_\alpha^{(1)} - \underline{\bar{x}}_1)(\underline{x}_\alpha^{(1)} - \underline{\bar{x}}_1)' \\ + \sum_{\alpha=1}^{N_2} (\underline{x}_\alpha^{(2)} - \underline{\bar{x}}_2)(\underline{x}_\alpha^{(2)} - \underline{\bar{x}}_2)'$$

But this doesn't have Wishart distribution with covariance matrix multiplied of $\frac{\Sigma_1}{N_1} + \frac{\Sigma_2}{N_2}$

$$\text{Now, } \underline{y} \sim N(\underline{\mu} - \underline{\mu}_2, \Sigma^*)$$

If $\Sigma_1 = \Sigma_2$ then the test statistic will follow Hotelling T^2

Case-I $N_1 = N_2 = N$

$$H_0: \underline{\mu}_1 = \underline{\mu}_2 \text{ or } \underline{\mu}_1 - \underline{\mu}_2 = \underline{\mu}^* = 0$$

$$\text{Then } \underline{\bar{X}}_1 - \underline{\bar{X}}_2 \sim N(\underline{\mu}_1 - \underline{\mu}_2, \frac{\Sigma_1}{N} + \frac{\Sigma_2}{N})$$

$$\text{Define, } \underline{Y}_\alpha = \underline{X}_\alpha^{(1)} - \underline{X}_\alpha^{(2)} \rightarrow \underline{\mu}^* \rightarrow \underline{\Sigma}^*$$

$$\sim N(\underline{\mu}_1 - \underline{\mu}_2, \Sigma_1 + \Sigma_2)$$

independent of $\underline{Y}_\beta (\beta \neq \alpha)$

$$\text{Let, } \underline{\bar{Y}} = \frac{1}{N} \sum_{\alpha=1}^N \underline{Y}_\alpha, \quad \underline{S} = \frac{1}{N-1} \sum_{\alpha=1}^N (\underline{Y}_\alpha - \underline{\bar{Y}})(\underline{Y}_\alpha - \underline{\bar{Y}})'$$

$$\text{Therefore, } \boxed{T^2 = N \underline{\bar{Y}}' \underline{S}^{-1} \underline{\bar{Y}}}$$

$$\underline{T}^2 \sim \frac{T^2}{N-1}$$

But in ~~the~~ Fisher-T - setup of univariate we had $n_1 + n_2 - 2$ d.f. . Similarly, we ~~to~~ should have $N + N - 2 \equiv 2(N-1)$ d.f. But we are having $N-1$ d.f. i.e. there is a loss of $N-1$ d.f.

[t-distn ^{with} higher df $\Rightarrow$ thinner tail]

$\Rightarrow$ less type-I error

So, here, probability of rejecting true H_0 becomes higher.]

Case-II $N_1 \neq N_2$

$$\underline{Y}_\alpha = \left(\underline{X}_\alpha^{(1)} - \sqrt{\frac{N_1}{N_2}} \underline{X}_\alpha^{(2)} \right) + \frac{1}{\sqrt{N_1 N_2}} \left(\sum_{\beta=1}^{N_1} \underline{X}_\beta^{(1)} - \sqrt{\frac{N_1}{N_2}} \sum_{\gamma=1}^{N_2} \underline{X}_\gamma^{(2)} \right)$$

$$= \left(\underline{X}_\alpha^{(1)} - \sqrt{\frac{N_1}{N_2}} \underline{X}_\alpha^{(2)} \right) + \left(\frac{1}{\sqrt{N_1 N_2}} \sum_{\beta=1}^{N_1} \underline{X}_\beta^{(1)} - \frac{1}{\sqrt{N_2}} \sum_{\gamma=1}^{N_2} \underline{X}_\gamma^{(2)} \right)$$

$$E(\underline{Y}_\alpha) = \underline{\mu}_1 - \sqrt{\frac{N_1}{N_2}} \underline{\mu}_2 + \frac{N_1}{\sqrt{N_1 N_2}} \underline{\mu}_1 - \frac{N_2}{\sqrt{N_1 N_2}} \underline{\mu}_2$$

$$= \underline{\mu}_1 - \underline{\mu}_2 = \underline{\mu}^*$$

$$\begin{aligned} \text{Cov}(\underline{Y}_\alpha) &= E[(\underline{Y}_\alpha - E(\underline{Y}_\alpha))(\underline{Y}_\alpha - E(\underline{Y}_\alpha))'] \\ &= \left(\Sigma_1 + \frac{N_1}{N_2} \Sigma_2 \right) \delta_{\alpha\beta} \quad \text{where } \delta_{\alpha\beta} = \begin{cases} 1, & \text{if } \alpha = \beta \\ 0, & \text{otherwise} \end{cases} \\ &= \Sigma^* \end{aligned}$$

So, $H_0: E(\underline{Y}_\alpha) = \underline{0}$

The test statistic is -

$$T^2 = N_1 \bar{\underline{Y}}' S^{-1} \bar{\underline{Y}} \sim T_{N-1}^2 \text{ (under } H_0 \text{)}$$

where $\bar{\underline{Y}} = \frac{1}{N_1} \sum_{\alpha=1}^{N_1} \underline{Y}_\alpha$

$$\bar{\underline{Y}} = \bar{\underline{X}}_1 - \bar{\underline{X}}_2$$

$$(N_1 - 1) S = \sum_{\alpha=1}^{N_1} (\underline{Y}_\alpha - \bar{\underline{Y}})(\underline{Y}_\alpha - \bar{\underline{Y}})'$$

Let's take a more generalized variance,

$$H_0: \sum_{i=1}^q \beta_i \underline{\mu}_i = \underline{0}$$

where $\beta_1, \dots, \beta_q$ are given scalars.

$$\underline{X}_\alpha^{(i)} \sim N(\underline{\mu}_i, \Sigma_i), \quad i = 1, 2, \dots, q$$

$$\alpha = (i-1)N_i + j$$

$$\underline{Y}_\alpha = \beta_1 \underline{X}_\alpha^{(1)} + \sum_{i=2}^q \beta_i \left(\underline{X}_\alpha^{(i)} - \frac{1}{N_i} \sum_{j=1}^{N_i} \underline{X}_j^{(i)} \right) + \frac{1}{N_1 N_2} \sum_{j=1}^{N_1} \underline{X}_j^{(1)}$$

$$E(\underline{Y}_\alpha) = \sum \beta_i \underline{\mu}_i$$

$$V(\underline{Y}_\alpha) = \left(\sum_{i=1}^q \frac{\beta_i^2 N_i}{N_1} \Sigma_i \right) \delta_{\alpha\beta}$$

Under H_0 , test statistic will follow -

DISTRIBUTION OF SAMPLE CORRELATION COEFFICIENTS

KARL PEARSON CORRELATION COEFFICIENT

(in bivariate sample)

$$r = \frac{\sum (x_i - \bar{x})(y_i - \bar{y})}{\sqrt{\sum (x_i - \bar{x})^2 \sum (y_i - \bar{y})^2}} = \frac{a_{12}}{\sqrt{a_{11} a_{22}}}$$

$$f(a_{11}, a_{12}, a_{22}) \equiv f(a_{11}, a_{12}, \sqrt{a_{11} a_{22}})$$

This transformation will serve the purpose.

PDF of r can be obtained by integrating out a_{11} & a_{22} .

Now, we know the PDF of Wishart distribution is —

$$f(A) = \frac{|A|^{n-p-1/2} \exp(-\frac{1}{2} \text{tr} \Sigma^{-1} A)}{2^{np/2} \prod_{i=1}^{p-1} \frac{\pi^{(n-i)/2}}{\Gamma(\frac{n-i}{2})} \pi^{p(p-1)/4} |\Sigma|^{n/2}}$$

$p=2$

$$A = \begin{pmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{pmatrix}, \quad \Sigma = \begin{pmatrix} \sigma_{11} & \sigma_{12} \\ \sigma_{21} & \sigma_{22} \end{pmatrix}, \quad \underline{x} = \begin{pmatrix} x_1 \\ x_2 \end{pmatrix}$$

~~$$\Sigma = \begin{pmatrix} \sigma_{11} & \sigma_{12} \\ \sigma_{12} & \sigma_{22} \end{pmatrix}$$~~

Now, $f(A) = f(a_{11}, a_{22}, a_{12})$.

$$|A| = a_{11} a_{22} - a_{12}^2$$

$$|\Sigma| = \sigma_1^2 \sigma_2^2 (1 - r^2)$$

$$\therefore |\Sigma^{-1}| = \frac{1}{\sigma_1^2 \sigma_2^2 (1 - r^2)}$$

r : Pearson's correlation coefficient.

$$\Sigma^{-1} A = \frac{1}{(1 - r^2)} \begin{pmatrix} 1/\sigma_1^2 & -r/\sigma_1 \sigma_2 \\ -r/\sigma_1 \sigma_2 & 1/\sigma_2^2 \end{pmatrix} \begin{pmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{pmatrix}$$

$$= \frac{1}{1-\rho^2} \begin{pmatrix} \frac{a_{11}}{\sigma_1^2} - \frac{a_{21}\rho}{\sigma_1\sigma_2} & \frac{a_{21}}{\sigma_1^2} - \frac{\rho a_{22}}{\sigma_2^2} \\ -\frac{a_{11}\rho}{\sigma_1\sigma_2} + \frac{a_{21}}{\sigma_2^2} & -\frac{a_{22}\rho}{\sigma_1\sigma_2} + \frac{a_{12}}{\sigma_2^2} \end{pmatrix}$$

$$\det(\Sigma^{-1}A) = \frac{1}{1-\rho^2} \left[\frac{a_{11}}{\sigma_1^2} + \frac{a_{22}}{\sigma_2^2} - \frac{2a_{12}\rho}{\sigma_1\sigma_2} \right]$$

$$= 8.$$

Back to wiswot PDF for $p=2$,

$$f(A) = \frac{(a_{11}a_{22} - a_{12}^2)^{\frac{n-3}{2}} \exp(-8/2)}{2^n \sqrt{\pi} \left[\frac{n}{2} \right] \left[\frac{n-1}{2} \right] |\Sigma|^{\frac{n}{2}}}$$

$$a_{12} = h \sqrt{a_{11}a_{22}}$$

$$da_{12} = \sqrt{a_{11}a_{22}} dh$$

$$f(A) = \frac{(a_{11}(a_{22} - h^2 a_{11}a_{22}))^{\frac{n-3}{2}}}{2^n \sqrt{\pi} \cdot \left[\frac{n}{2} \right] \left[\frac{n-1}{2} \right] |\Sigma|^{\frac{n}{2}}}$$

$$= \frac{1}{2^n \sqrt{\pi} \cdot \left[\frac{n}{2} \right] \left[\frac{n-1}{2} \right] |\Sigma|^{\frac{n}{2}}} \cdot \exp \left[-\frac{1}{2(1-\rho^2)} \left\{ \frac{a_{11}}{\sigma_1^2} + \frac{a_{22}}{\sigma_2^2} - \frac{2\rho h \sqrt{a_{11}a_{22}}}{\sigma_1\sigma_2} \right\} \right] |J|$$

$$= L^* \exp \left[-\frac{1}{2(1-\rho^2)} \frac{a_{11}}{\sigma_1^2} \right] \exp \left[-\frac{1}{2(1-\rho^2)} \frac{a_{22}}{\sigma_2^2} \right]$$

$$\exp \left[\frac{\rho h}{\sigma_1\sigma_2} \frac{\sqrt{a_{11}a_{22}}}{(1-\rho^2)} \right]$$

$$= \frac{a_{11}^{n/2-1} a_{22}^{n/2-1} (1-r^2)^{\frac{n-3}{2}}}{2^n |\Sigma|^{n/2} \sqrt{\pi} \Gamma\left(\frac{n}{2}\right) \Gamma\left(\frac{n-1}{2}\right)} \exp\left(-\frac{a_{11}}{2\sigma_1^2(1-r^2)}\right) \exp\left(-\frac{a_{22}}{2\sigma_2^2(1-r^2)}\right) \left[\sum_{\alpha=0}^{\infty} \left(\frac{f(r) \sqrt{a_{11}a_{22}}}{\sigma_1\sigma_2(1-r^2)} \right)^{\alpha} \frac{1}{\alpha!} \right]$$

[Taylor series expansion of the 3rd term]

$$= \frac{(1-r^2)^{\frac{n-3}{2}}}{2^n \Gamma\left(\frac{n}{2}\right) \Gamma\left(\frac{n-1}{2}\right) \sqrt{\pi} |\Sigma|^{n/2}} \sum_{\alpha=0}^{\infty} \left[\left\{ a_{11}^{\frac{n+\alpha}{2}-1} \exp\left(-\frac{a_{11}}{2(1-r^2)\sigma_1^2}\right) \right\} \left\{ a_{22}^{\frac{n+\alpha}{2}-1} \exp\left(-\frac{a_{22}}{2(1-r^2)\sigma_2^2}\right) \right\} \left\{ \left(\frac{f(r)}{\sigma_1\sigma_2(1-r^2)} \right)^{\alpha} \cdot \frac{1}{\alpha!} \right\} \right]$$

The PDF of x is then =

$$\therefore f(x) = \int_0^{\infty} \int_0^{\infty} f(a_{11}, a_{22}, x) da_{11} da_{22}$$

Now, $\int_0^{\infty} a_{11}^{\frac{n+\alpha}{2}-1} \exp\left(-\frac{a_{11}}{2(1-r^2)\sigma_1^2}\right) da_{11}$

$$= \left[\frac{a_{11}^{\frac{n+\alpha}{2}}}{\frac{n+\alpha}{2}} \right]_0^{\infty} \left[2\sigma_1^2(1-r^2) \right]^{\frac{n+\alpha}{2}}$$

$$\int_0^{\infty} a_{22}^{\frac{n+\alpha}{2}-1} \exp\left(-\frac{a_{22}}{2(1-r^2)\sigma_2^2}\right) da_{22}$$

$$= \left[\frac{a_{22}^{\frac{n+\alpha}{2}}}{\frac{n+\alpha}{2}} \right]_0^{\infty} \left[2\sigma_2^2(1-r^2) \right]^{\frac{n+\alpha}{2}}$$

$$\therefore f(x) = \frac{(1-r^2)^{n-3/2}}{2^{n/2} \Gamma\left(\frac{n}{2}\right) \Gamma\left(\frac{n-1}{2}\right) \sqrt{\pi} \sigma_1^2 \sigma_2^2 (1-r^2)} \sum_{\alpha=0}^{\infty} \left[\frac{f(x)^{\alpha}}{\left(\frac{\sigma_1^{\alpha} \sigma_2^{\alpha} (1-r^2)^{\alpha}}{\alpha!} \right)^{\alpha}} \right]$$

$$= \left[\left(\frac{n+\alpha}{2} \right)^2 2^{n+\alpha} \sigma_1^{n+\alpha} \sigma_2^{n+\alpha} (1-r^2)^{n+\alpha} \right]$$

$$= \frac{(1-x^2)^{\frac{n-3}{2}} 2^n \sigma_1^n \sigma_2^n (1-y^2)^n}{2^n \sqrt{\frac{n}{2}} \sqrt{\frac{n-1}{2}} \sqrt{\pi} (\sigma_1^2 \sigma_2^2 (1-y^2))^{\frac{n}{2}}}$$

$$\sum_{\alpha=0}^{\infty} \left[\frac{\left(\sqrt{\frac{n+\alpha}{2}}\right)^2 (y^2)^{\alpha} 2^{\alpha} \sigma_1^{\alpha} \sigma_2^{\alpha} (1-y^2)^{\alpha}}{\sigma_1^{\alpha} \sigma_2^{\alpha} (1-y^2)^{\alpha} \alpha!} \right]$$

$$= \frac{(1-x^2)^{\frac{n-3}{2}} (1-y^2)^{\frac{n}{2}}}{\sqrt{\frac{n}{2}} \sqrt{\frac{n-1}{2}} \sqrt{\pi}} \sum_{\alpha=0}^{\infty} \frac{(2y^2)^{\alpha}}{\alpha!} \left(\sqrt{\frac{n+\alpha}{2}}\right)^2$$

Now, By duplication formula,

$$\Gamma(m) \Gamma(m+\frac{1}{2}) = \frac{\sqrt{\pi}}{2^{2m-1}} \Gamma(2m)$$

$$\text{Put } m = \frac{n-1}{2} \quad \Gamma\left(\frac{n-1}{2}\right) \Gamma\left(\frac{n-1}{2} + \frac{1}{2}\right) = \frac{\sqrt{\pi} \Gamma(n-1)}{2^{n-2}}$$

$$= \frac{(1-x^2)^{\frac{n-3}{2}} (1-y^2)^{\frac{n}{2}} 2^{n-2}}{\sqrt{\pi} \sqrt{\pi} \Gamma(n-1)} \sum_{\alpha=0}^{\infty} \left[\frac{(2y^2)^{\alpha}}{\alpha!} \left(\sqrt{\frac{n+\alpha}{2}}\right)^2 \right]$$

$$= \frac{2^{n-2} (1-x^2)^{\frac{n-3}{2}} (1-y^2)^{\frac{n}{2}}}{\pi (n-2)!} \sum_{\alpha=0}^{\infty} \left[\frac{(2y^2)^{\alpha}}{\alpha!} \left(\sqrt{\frac{n+\alpha}{2}}\right)^2 \right]$$

Here, $n = N+1$

The only function of f which can be estimated unbiasedly is $(\sin^{-1} f)$

$$E(\sin^{-1} h) = \sin^{-1} f.$$

$$[\text{But, } E(h) \neq f]$$

$$\left. \begin{aligned} E(h) &\approx f - \frac{f(1-f^2)}{2n} + O\left(\frac{1}{n^2}\right) \\ \text{and } \text{Var}(h) &\approx \frac{(1-f^2)^2}{n} \left(1 + \frac{11f^2}{2n}\right) + O\left(\frac{1}{n^3}\right) \end{aligned} \right\}$$

Transform f to $z = \frac{1}{2} \log_e \frac{1+f}{1-f}$

Convergence towards Normal distribution is more rapid for z transformation.

$$P(h \leq h^*) = F(h^* | N, f) \rightarrow \text{These values are tabulated for } h^* = 1, -0.95$$

12/02/2024

DISTRIBUTION OF CORRELATION COEFFICIENT UNDER $f=0$.

$$f(h) = \frac{(1-h^2)^{n-3/2}}{\Gamma_{n/2} \Gamma_{n-1/2} \sqrt{\pi}} \sum_{\alpha=0}^{\infty} \frac{(2f h)^{\alpha}}{\alpha!} \left(\frac{n+\alpha}{2} \right)^{\alpha}$$

when $f=0$,

$$= \frac{(1-h^2)^{n-3/2}}{\Gamma_{n/2} \Gamma_{n-1/2} \sqrt{\pi}} \left(\frac{n}{2} \right)^n \left\{ \begin{aligned} &\alpha=0 \text{ (case)} + \sum_{\alpha=1}^{\infty} \dots \\ &= \frac{(2f h)^0}{0!} + \left(\frac{n+0}{2} \right)^2 + \dots \end{aligned} \right.$$

Now, we know, $\sqrt{\pi} = \Gamma_{1/2}$

$$f(h) |_{f=0} = (1-h^2)^{\frac{n-3}{2}} \frac{\Gamma_{n/2}}{\Gamma_{n-1/2} \Gamma_{1/2}} (1-h^2)$$

$$= \frac{(1-r^2)^{N-1/2}}{B(1/2, \frac{N-2}{2})}$$

$$= \text{constant} \cdot (1-r^2)^{\frac{N-1}{2}}$$

↪ (Symmetric)

Alt

$$r = \frac{\sum_{i=1}^n (x_{1i} - \bar{x}_1)(x_{2i} - \bar{x}_2)}{\sqrt{\sum_{i=1}^n (x_{1i} - \bar{x}_1)^2 \sum_{i=1}^n (x_{2i} - \bar{x}_2)^2}}$$

Say, $f(x_1, x_2) \sim N(0, 0, \sigma_1^2, \sigma_2^2, 0)$

Consider linear orthogonal transformation

$$Y = CX, \quad CC' = I$$

$$\begin{pmatrix} y_1 \\ \vdots \\ y_n \end{pmatrix} = \begin{pmatrix} c_{11} & \dots & c_{1n} \\ \vdots & \ddots & \vdots \\ c_{n1} & \dots & c_{nn} \end{pmatrix} \begin{pmatrix} x_1 \\ \vdots \\ x_n \end{pmatrix}$$

$$\Rightarrow y_i = \sum_{\alpha=1}^n c_{i\alpha} x_{\alpha}$$

$$\text{choose, } c_{1\alpha} = \frac{1}{\sqrt{n}}, \quad y_1 = \frac{1}{\sqrt{n}}(x_{11} + x_{12} + \dots + x_{1n}) = \sqrt{n} \bar{x}_1$$

$$\text{Now, } Y'Y = X'X$$

$$\begin{aligned} \text{Term from } X'X &\rightarrow \sum_{\alpha=1}^n x_{1\alpha}^2 = \sum_{\alpha=1}^n [(x_{1\alpha} - \bar{x}_1) + \bar{x}_1]^2 \\ &= \sum_{\alpha=1}^n (x_{1\alpha} - \bar{x}_1)^2 + \underbrace{n \bar{x}_1^2}_{y_1^2} \end{aligned}$$

$$\text{Term from } Y'Y \rightarrow \sum_{\alpha=1}^n y_{\alpha}^2 = \sum_{\alpha=1}^n \left(\frac{1}{\sqrt{n}} \right)^2 + y_1^2$$

$$\sum_{\alpha=1}^n y_{1\alpha}^2 - y_1^2 = \sum_{\alpha=2}^n (x_{1\alpha} - \bar{x}_1)^2 \quad y_i \sim N(0, \sigma_1^2)$$

$$= \sum_{\alpha=2}^n (x_{1\alpha} - \bar{x}_1)^2 \quad i = 2(1)n$$

$$\boxed{\frac{\sum_{\alpha=2}^n y_{1\alpha}^2}{\sigma^2} \sim \chi^2_{n-1}}$$

$$E(y_i) = \sum c_{i\alpha} E(x_{1\alpha})$$

$$= 0$$

$$E(y_1^2) = \sum c_{1\alpha}^2 E(x_{1\alpha}^2)$$

$$+ \sum_{i \neq j} \sum c_{1\alpha} c_{1\beta} E(x_{1\alpha} x_{1\beta})$$

$$= \sum c_{1\alpha}^2 \sigma_1^2 + 0$$

$$= \sigma_1^2$$

$$y_2 = \sum_{\alpha=2}^n c_{2\alpha} x_{1\alpha}$$

$$= \sum_{\alpha=1}^n \frac{(x_{2\alpha} - \bar{x}_2)}{\sqrt{\sum (x_{2\alpha} - \bar{x}_2)^2}} x_{1\alpha}$$

$$= \sum_{\alpha=1}^n \frac{(x_{2\alpha} - \bar{x}_2)(x_{1\alpha} - \bar{x}_1)}{\sqrt{\sum (x_{2\alpha} - \bar{x}_2)^2}} + 0$$

$$= r_2 \sqrt{\sum (x_{1\alpha} - \bar{x}_1)^2}$$

$$= r_2 \sqrt{\sum_{\alpha=2}^n y_{1\alpha}^2}$$

$$\sum_{\alpha=3}^n y_{1\alpha}^2 = \sum_{\alpha=2}^n y_{1\alpha}^2 - y_2^2 = (1 - r_2^2) \sum_{\alpha=2}^n (x_{1\alpha} - \bar{x}_1)^2$$

independent

$$\left\{ \begin{aligned} \frac{\sum_{\alpha=3}^n y_{1\alpha}^2}{\sigma_1^2} &= \frac{(1 - r_2^2) \sum (x_{1\alpha} - \bar{x}_1)^2}{\sigma_1^2} \sim \chi^2_{n-2} \\ \frac{y_2^2}{\sigma_1^2} &= \frac{r_2^2 \sum (x_{1\alpha} - \bar{x}_1)^2}{\sigma_1^2} \sim \chi^2_1 \end{aligned} \right.$$

$$r^2 = \frac{\chi^2(1)}{\chi^2(1) + \chi^2(n-2)} \sim \text{Beta}\left(\frac{1}{2}, \frac{n-2}{2}\right)$$

$$f(r^2) dr^2 = \frac{1}{B\left(\frac{1}{2}, \frac{n-2}{2}\right)} (r^2)^{-1/2} (1-r^2)^{\frac{n-4}{2}} dr^2$$

$$dr^2 = 2r dr$$

$$f(r) dr = \frac{1}{B\left(\frac{1}{2}, \frac{n-2}{2}\right)} (1-r^2)^{\frac{n-4}{2}} dr$$

26/02/2024

DISTRIBUTION OF SET OF CORRELATION COEFFICIENT FOR A DIAGONAL POPULATION MATRIX

<> Find the density of the set r_{ij} ($i < j$, $i, j = 1, \dots, p$) where each $\rho_{ij} = 0$ ($i \neq j$).

$$f(A) = \frac{|A|^{\frac{n-p-1}{2}} \exp\left(-\frac{1}{2} A \Sigma^{-1} A\right)}{2^{\frac{np}{2}} \prod_{i=1}^{p-1} \left(\frac{\pi}{2}\right)^{\frac{n-i}{2}} \left(\prod_{i=1}^{p-1} \sigma_{ii}\right)^{\frac{n}{2}}}$$

$$|\Sigma| = \begin{vmatrix} \sigma_{11} & 0 \\ 0 & \sigma_{pp} \end{vmatrix} = \prod_{i=1}^p \sigma_{ii}$$

$$f(A) = \frac{|A|^{\frac{n-p-1}{2}} \exp\left(-\frac{1}{2} \sum \frac{a_{ij}^2}{\sigma_{ii}}\right)}{2^{\frac{np}{2}} \prod_{i=1}^{p-1} \left(\frac{\pi}{2}\right)^{\frac{n-i}{2}} \prod_{i=1}^p \sigma_{ii}^{\frac{n}{2}}}$$

Now, $a_{ij} = r_{ij} \sqrt{a_{ii} a_{jj}}$ ($i \neq j$) $f(r, s) = f(A) |J|$

$$\frac{\partial a_{ij}}{\partial r_{ij}} = \sqrt{a_{ii} a_{jj}}, \quad \frac{\partial a_{ij}}{\partial a_{ii}} = \frac{1}{2} \sqrt{\frac{a_{jj}}{a_{ii}}} r_{ij}$$

$$|J| = \frac{\partial(a_{11}, \dots, a_{pp}, a_{12}, \dots, a_{1p}, a_{23}, \dots, a_{2p}, \dots, a_{p-1,p})}{\partial(a_{11}, a_{22}, \dots, a_{pp}, \kappa_{12}, \dots, \kappa_{1p}, \pi_{23}, \dots, \pi_{2p}, \dots, \kappa_{p-1,p})}$$

$$= \begin{vmatrix} \frac{\partial a_{11}}{\partial a_{11}} & \frac{\partial a_{22}}{\partial a_{11}} & \dots & \frac{\partial a_{pp}}{\partial a_{11}} & \frac{\partial a_{12}}{\partial a_{11}} & \dots & \frac{\partial a_{1p}}{\partial a_{11}} & \dots & \frac{\partial a_{p-1,p}}{\partial a_{11}} \\ \frac{\partial a_{11}}{\partial a_{22}} & \frac{\partial a_{22}}{\partial a_{22}} & \dots & \frac{\partial a_{pp}}{\partial a_{22}} & \frac{\partial a_{12}}{\partial a_{22}} & \dots & \frac{\partial a_{1p}}{\partial a_{22}} & \dots & \frac{\partial a_{p-1,p}}{\partial a_{22}} \\ \vdots & \vdots & \ddots & \vdots & \vdots & \ddots & \vdots & \ddots & \vdots \\ \frac{\partial a_{11}}{\partial a_{pp}} & \frac{\partial a_{22}}{\partial a_{pp}} & \dots & \frac{\partial a_{pp}}{\partial a_{pp}} & \frac{\partial a_{12}}{\partial a_{pp}} & \dots & \frac{\partial a_{1p}}{\partial a_{pp}} & \dots & \frac{\partial a_{p-1,p}}{\partial a_{pp}} \\ \frac{\partial a_{11}}{\partial \kappa_{12}} & \frac{\partial a_{22}}{\partial \kappa_{12}} & \dots & \frac{\partial a_{pp}}{\partial \kappa_{12}} & \frac{\partial a_{12}}{\partial \kappa_{12}} & \dots & \frac{\partial a_{1p}}{\partial \kappa_{12}} & \dots & \frac{\partial a_{p-1,p}}{\partial \kappa_{12}} \\ \vdots & \vdots & \ddots & \vdots & \vdots & \ddots & \vdots & \ddots & \vdots \\ \frac{\partial a_{11}}{\partial \kappa_{p-1,p}} & \frac{\partial a_{22}}{\partial \kappa_{p-1,p}} & \dots & \frac{\partial a_{pp}}{\partial \kappa_{p-1,p}} & \frac{\partial a_{12}}{\partial \kappa_{p-1,p}} & \dots & \frac{\partial a_{1p}}{\partial \kappa_{p-1,p}} & \dots & \frac{\partial a_{p-1,p}}{\partial \kappa_{p-1,p}} \end{vmatrix}$$

$$= \begin{vmatrix} 1 & 0 & \dots & 0 & \frac{1}{2} \sqrt{\frac{a_{22}}{a_{11}}} \kappa_{12} & \dots & \frac{1}{2} \sqrt{\frac{a_{pp}}{a_{11}}} & 0 & \dots & 0 \\ 0 & 1 & \dots & 0 & \frac{1}{2} \sqrt{\frac{a_{11}}{a_{22}}} \kappa_{12} & \dots & 0 & \dots & 0 \\ \vdots & \vdots & \ddots & \vdots & \vdots & \ddots & \vdots & \ddots & \vdots \\ 0 & \dots & 0 & \sqrt{a_{11} a_{22}} & \dots & \dots & \sqrt{a_{p-1,p-1} a_{pp}} \end{vmatrix}$$

$$= \prod_{i=1}^p a_{ii}^{\frac{p-1}{2}}$$

$$|A| = \begin{vmatrix} a_{11} & \sqrt{a_{11}a_{22}}r_{12} \\ & a_{22} \end{vmatrix}$$

$$= \prod_{i=1}^p a_{ii} \begin{vmatrix} 1 & r_{12} & r_{1p} \\ & \ddots & \vdots \\ & & 1 \end{vmatrix} = \prod_{i=1}^p a_{ii} |R|$$

where $R = (r_{ij})$

$$f(a_{11}, \dots, a_{pp}, r_{ij}) = f(A) |J|$$

$$= |R|^{\frac{n-p-1}{2}} \left(\prod_{i=1}^p a_{ii} \right)^{\frac{n-p-1}{2}} \exp\left(-\frac{1}{2} \sum_{i=1}^p \frac{a_{ii}}{\sigma_{ii}}\right) \prod_{i=1}^p a_{ii}^{\frac{p-1}{2}}$$

$$= |R|^{n-p-1/2} \prod_{i=1}^p \left(\frac{a_{ii}^{n/2-1} \exp\left(-\frac{1}{2} \frac{a_{ii}}{\sigma_{ii}}\right)}{2^{n/2} \sigma_{ii}^{n/2}} \right)$$

$$f(r_{ij}) = \text{integral density } f(A) |J|$$

let, $\frac{a_{ii}}{2\sigma_{ii}} = u_i$

$$\int_0^\infty a_{ii}^{n/2-1} \exp\left(-\frac{1}{2} \frac{a_{ii}}{\sigma_{ii}}\right) da_{ii}$$

$$= 2^{n/2} \sigma_{ii} \int_0^\infty u_i^{n/2-1} e^{-u_i} du_i$$

$$= 2^{n/2} \sigma_{ii}^{n/2} \Gamma(n/2)$$

For (i,j) , $i,j=1, \dots, p$.

$f(a_{ij})$ = Marginal density from $f(A|J)$

$$= \int_0^\infty \dots \int_0^\infty \frac{|R|^{n-p-1/2}}{\prod_{i=0}^{p-1} \Gamma(\frac{n-i}{2}) \pi^{\frac{p(p-1)}{4}}} \prod_{i=1}^p (a_{ii})^{\frac{p-1}{2}} da_{11} \dots da_{pp}$$

$$\frac{a_{ii}}{\sigma_{ii}} \sim \chi^2_n$$

$$f(a_{ij}) = \frac{|R|^{n-p-1/2}}{\prod_{i=1}^p \sigma_{ii}^{n/2} 2^{np/2} \pi^{p(p-1)/4} \prod_{i=0}^{p-1} \Gamma(\frac{n-i}{2})} \prod_{i=1}^p \exp(-\frac{1}{2} \frac{a_{ii}}{\sigma_{ii}}) da_{ii}$$

$$= \frac{|R|^{n-p-1/2}}{\pi^{p(p-1)/4} \prod_{i=0}^{p-1} \Gamma(\frac{n-i}{2})}$$

e.g. 7 $\left(\begin{matrix} p=2 \\ \sigma_{ii} \end{matrix} \right) = \bar{X} = \frac{1}{N} \left(\begin{matrix} -SA & 1A \\ SA & 1A \end{matrix} \right) \frac{1}{N} = \frac{1}{N} \frac{1}{1-k^2} \frac{1}{2}$

$$f(a_{ij}) = \frac{1}{N} \frac{1}{1-k^2} \frac{1}{2} \text{Beta}(\frac{1}{2}, \frac{N-2}{2}) \left(\frac{a_{ii}}{N} \right) = \frac{1}{N} \frac{1}{1-k^2} \frac{1}{2}$$

For $N=2$?

For $N=5, p=2$, if $P(|R| \geq k) = \alpha$

$$k \sqrt{1-k^2} + \sin^{-1} k = \frac{\pi(1-\alpha)}{2}$$

$$\left(\frac{1}{N} \frac{1}{1-k^2} \frac{1}{2} \right) \frac{1}{N} = \frac{1}{N} \frac{1}{1-k^2} \frac{1}{2}$$

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DISTRIBUTION OF PARTIAL CORRELATION COEFFICIENT

$$\underline{X} \sim N(\underline{\mu}, \underline{\Sigma})$$

$$\underline{X} = \begin{pmatrix} X^{(1)} \\ X^{(2)} \end{pmatrix}, \quad \underline{\mu} = \begin{pmatrix} \mu_1 \\ \mu_2 \end{pmatrix}$$

$$\underline{\Sigma} = \begin{pmatrix} \Sigma_{11} & \Sigma_{12} \\ \Sigma_{21} & \Sigma_{22} \end{pmatrix}$$

$$X^{(1)} | X^{(2)} = x^{(2)} \sim N(\mu^{(1)} + \beta(x^{(2)} - \mu^{(2)}), \Sigma_{11.2})$$

$$\Sigma_{11.2} = \Sigma_{11} - \Sigma_{12} \Sigma_{22}^{-1} \Sigma_{21}$$

$$\beta = \Sigma_{12} \Sigma_{22}^{-1}$$

Partial correlation coefficient w/o x_i & x_j [$x_i, x_j \in X$]
holding $x_{q+1}, \dots, x_p$ (i.e. x_2) as fixed.

$$= \rho_{ij \cdot 2 \dots p}$$

Now, $\hat{\Sigma} = \frac{A}{N} = \frac{1}{N} \begin{pmatrix} A_{11} & A_{12} \\ A_{21} & A_{22} \end{pmatrix}, \quad \hat{\mu} = \bar{X} = \begin{pmatrix} \bar{X}^{(1)} \\ \bar{X}^{(2)} \end{pmatrix}$

If $\Sigma = \begin{pmatrix} \Sigma_{11} & \Sigma_{12} \\ \Sigma_{21} & \Sigma_{22} \end{pmatrix}$ is known, $\Sigma_{11.2}$ & β are also known.

If $(\Sigma_{22}, \beta, \Sigma_{11.2})$ are known, then $\Sigma_{12} = \beta \Sigma_{22}$.

& $\Sigma_{11} = \Sigma_{11.2} + \beta' \Sigma_{22} \beta$

Theorem $\hat{\theta}$ is MLE of θ .

Then $f(\hat{\theta})$ is MLE of $f(\theta)$.

If f is a one-one fn.

Here,

$$\hat{\mu}^{(1)} = \bar{X}^{(1)}, \quad \mu^{(2)} = \bar{X}^{(2)}$$

$$\hat{\beta} = A_{12} A_{22}^{-1} \quad \& \quad \hat{\Sigma}_{11.2} = \frac{1}{N} (A_{11} - A_{12} A_{22}^{-1} A_{21})$$

$$\rho_{ij, \bar{q}+1, \dots, p} = \frac{\sigma_{ij, \bar{q}+1, \dots, p}}{\sqrt{\sigma_{ii, \bar{q}+1, \dots, p} \cdot \sigma_{jj, \bar{q}+1, \dots, p}}}$$

Then MLE of $\rho_{ij, \bar{q}+1, \dots, p}$ is -

$$\hat{\rho}_{ij, \bar{q}+1, \dots, p} = \frac{\hat{\sigma}_{ij, \bar{q}+1, \dots, p}}{\sqrt{\hat{\sigma}_{ii, \bar{q}+1, \dots, p} \cdot \hat{\sigma}_{jj, \bar{q}+1, \dots, p}}}$$

$$= \frac{(a_{ij, \bar{q}+1, \dots, p})^{(1)} \dots (a_{ij, \bar{q}+1, \dots, p})^{(n)}}{\sqrt{(a_{ii, \bar{q}+1, \dots, p})^{(1)} \dots (a_{ii, \bar{q}+1, \dots, p})^{(n)} \cdot (a_{jj, \bar{q}+1, \dots, p})^{(1)} \dots (a_{jj, \bar{q}+1, \dots, p})^{(n)}}}$$

$$= \frac{(a_{ij, \bar{q}+1, \dots, p})^{(1)} \dots (a_{ij, \bar{q}+1, \dots, p})^{(n)}}{\sqrt{(a_{ii, \bar{q}+1, \dots, p})^{(1)} \dots (a_{ii, \bar{q}+1, \dots, p})^{(n)} \cdot (a_{jj, \bar{q}+1, \dots, p})^{(1)} \dots (a_{jj, \bar{q}+1, \dots, p})^{(n)}}}$$

$A = \sum x_{\alpha} x_{\alpha}'$ and $\sigma_{ij, \bar{q}+1, \dots, p}$: (i, j) th element of $\Sigma_{11,2}$

$$\begin{aligned} A_{11,2} &= \sum (x_{\alpha 1} | x_{\alpha 2} = \bar{x}_2) (x_{\alpha 1} | x_{\alpha 2} = \bar{x}_2)' \\ &= \sum_{\alpha=1}^n \left[\left(x_{\alpha 1}^{(1)} - \bar{x}^{(1)} - \hat{\beta} (x_{\alpha 2}^{(2)} - \bar{x}^{(2)}) \right) \right. \\ &\quad \left. \left(x_{\alpha 1}^{(1)} - \bar{x}^{(1)} - \hat{\beta} (x_{\alpha 2}^{(2)} - \bar{x}^{(2)}) \right)' \right] \end{aligned}$$

(Here A is analogous to R)

Residuals

Residuals of $x_{\alpha}^{(1)}$ from its regressor $x_{\alpha}^{(2)}$

So, Partial correlation coefficient is simple correlation coefficient among the residuals mentioned above.

(i.e. simple correlation coefficient from $A_{11,2}$)

$$A = \sum x_{\alpha} x_{\alpha}' \stackrel{i.i.d.}{\sim} N(0, \Sigma)$$

$$A_{11,2} = \sum_{\alpha=1}^n (x_{\alpha 1} | x_{\alpha 2} = \bar{x}_2) (x_{\alpha 1} | x_{\alpha 2} = \bar{x}_2)' \sim W(\Sigma_{11,2}, n^*)$$

Now here's two questions -

- (i) $x_{\alpha 1} | x_{\alpha 2} \sim \text{ind} ?$
- (ii) $n^* = ?$

n by n^* .

$$\underline{y}_2 = \begin{pmatrix} y_2^{(1)} \\ y_2^{(2)} \end{pmatrix}, \quad \underline{y}_2 \sim f(y_2). \rightarrow \text{Marginal density}$$
$$f(\underline{y}_2^{(2)} | \underline{y}_2^{(1)} = \underline{y}_2^{(1)}) : \text{conditional density.}$$

Then in $f(y_1^{(2)}, \dots, y_n^{(2)} | y_1^{(1)} = y_1^{(1)}, \dots, y_n^{(1)} = y_n^{(1)})$.

$y_1^{(2)}, \dots, y_n^{(2)}$ are independent and

~~$$y_1 = x_1 + Bx_2$$~~
$$[\text{Recall } \left(\frac{1}{n} \sum_{i=1}^n \bar{X}_i - \frac{1}{n} \sum_{i=1}^n W(\Sigma_{ii}, 2, n-p+r) \bar{X}_i \right)]$$

Multiple Correlation Coefficient

Cauchy Schwarz Inequality

$(x'y)^2 \leq (x'x) \cdot (y'y)$ and $x'y = (x'x) \cdot (y'y)$
 $(x,0)$ and x only when.

$\underline{x} = \begin{pmatrix} x_1 \\ x_2 \\ \vdots \\ x_m \end{pmatrix} \quad \underline{x} = \begin{pmatrix} x_1 \\ x_2 \end{pmatrix}$

$$\Sigma = \begin{pmatrix} \Sigma_{11} & \Sigma_{11}' \\ \Sigma_{11} & \Sigma_{22} \end{pmatrix}$$

The multiple correlation coefficient b/w x_1 & $x_2, \dots, x_m$

$R_{1.23\dots m}$ is the maximum correlation b/w x_1 & any linear function $\alpha'x$ of $x_2, \dots, x_m$.

$$R_{1.23\dots m} = \max_{\alpha} \frac{\text{Cov}(x_1, \alpha'x_2)}{\sqrt{\text{Var}(x_1) \text{Var}(\alpha'x_2)}} \\ = \max_{\alpha} \frac{\alpha' \sigma_{c1}}{\sqrt{\sigma_{11} \alpha' \Sigma_{22} \alpha}}$$

Now,

$$\frac{\alpha' \sigma_{c1}}{\sqrt{\sigma_{11} \alpha' \Sigma_{22} \alpha}} = \frac{\alpha' \Sigma_{22}^{-1/2} \Sigma_{22}^{1/2} \sigma_{c1}}{\sqrt{\sigma_{11} \alpha' \Sigma_{22} \alpha}}$$

$$\left(\frac{u'v}{\sqrt{u'u} \sqrt{v'v}} \right) = \frac{u'v}{\sqrt{u'u} \sqrt{v'v}} = \frac{u' \Sigma_{22}^{-1/2} \alpha}{\sqrt{u'u} \sqrt{v'v}} = \frac{u' \Sigma_{22}^{-1/2} \alpha}{\sqrt{u'u} \sqrt{v'v}}$$

$$\sqrt{\sigma_{11} \alpha' \Sigma_{22} \alpha} = \sqrt{\sigma_{11} \alpha' \Sigma_{22} \alpha}$$

$$\frac{(\alpha' \Sigma_{22} \alpha)^{1/2} (\sigma_{c1}' \Sigma_{22}^{-1} \sigma_{c1})^{1/2}}{\sqrt{\sigma_{11} \alpha' \Sigma_{22} \alpha}}$$

$$= \frac{\sqrt{\sigma_{c1}' \Sigma_{22}^{-1} \sigma_{c1}}}{\sqrt{\sigma_{11}}}$$

equality holds when $\lambda u + \mu v = 0$

$$\lambda \Sigma_{22}^{-1/2} \alpha + \mu \Sigma_{22}^{1/2} \sigma_{c1} = 0$$

or, $\alpha = \Sigma_{22}^{-1} \sigma_{c1}$

$$R_{1.23\dots m} = \max_{\alpha} \frac{\alpha' \sigma_{c1}}{\sqrt{\sigma_{11} \alpha' \Sigma_{22} \alpha}} = \frac{\sigma_{c1}' \Sigma_{22}^{-1} \sigma_{c1}}{\sigma_{11}}$$

$$R^2_{1.23\dots m} = \frac{\sigma_{c1}' \Sigma_{22}^{-1} \sigma_{c1}}{\sigma_{11}} = R^2_{1.23\dots m}$$

We know $\Sigma_{11.2} = \Sigma_{11} - \Sigma_{12} \Sigma_{22}^{-1} \Sigma_{21}$

$$X = \begin{pmatrix} x_1 \\ x_2 \end{pmatrix} \rightarrow (p-q) \times 1$$

Take $q = 1$

$$\text{Then } \Sigma_{11.2} = \Sigma_{11} - \Sigma_{11} \Sigma_{22}^{-1} \Sigma_{11}$$

$$R_{1.23 \dots m} = \Sigma_{11} - \Sigma_{11}$$

$$R_{1.23 \dots m} = \Sigma_{11} - \Sigma_{11.2}$$

In general,

$$R_{1.23 \dots p} = \frac{\Sigma_{11} - \Sigma_{11.23 \dots p}}{\Sigma_{11}}$$

$$\text{Now } \Sigma_{11.2} = (\Sigma_{11} - \Sigma_{11.23 \dots p})$$

$$(\hat{\beta})' R = \sqrt{\frac{\Sigma_{11}' \Sigma_{22}^{-1} \Sigma_{11}}{\Sigma_{11}}} = \sqrt{\frac{\beta' \Sigma_{22}^{-1} \beta}{\Sigma_{11}}} \quad \text{where, } \beta = \Sigma_{22}^{-1} \Sigma_{11}$$

$$(\hat{R}) : R = \sqrt{\frac{\beta' \Sigma_{22}^{-1} \beta}{\Sigma_{11}}}$$

$$= \sqrt{\frac{\Sigma_{11} \Sigma_{22}^{-1} \Sigma_{11}}{\Sigma_{11}}} \quad \hat{\beta} = A_{22}^{-1} a_{21}$$

[Actually, All things are coming from the fact $\hat{\Sigma} = \frac{A}{N}$]

There is a one-one correspondence b/w $(\bar{R}, \Sigma_{11}, \Sigma_{22})$ & Σ .

$$\text{Say, } \Sigma = \begin{pmatrix} \Sigma_{11} & \Sigma_{12} \\ \Sigma_{21} & \Sigma_{22} \end{pmatrix} \text{ is given.}$$

$$\text{Then } R = \sqrt{\frac{\Sigma_{11}' \Sigma_{22}^{-1} \Sigma_{11}}{\Sigma_{11}}} \text{ is known}$$

$$R, \Sigma_{11}, \Sigma_{22} \text{ is known.}$$

$$\therefore \Sigma_{11} = \frac{\Sigma_{11}' \Sigma_{22}^{-1} \Sigma_{11}}{R^2}$$

So, R is the MLE of $\bar{R}$.

$$R^2 = \frac{\underline{a}_{(1)}' A_{22}^{-1} \underline{a}_{(1)}}{a_{11}}$$

$$1 - R^2 = \frac{a_{11} - \underline{a}_{(1)}' A_{22}^{-1} \underline{a}_{(1)}}{a_{11}} = \frac{a_{11.2}}{a_{11}}$$

$$\frac{R^2}{1 - R^2} = \frac{\underline{a}_{(1)}' A_{22}^{-1} \underline{a}_{(1)}}{a_{11.2}}$$

We know, $A_{11.2} \sim W(n - p + 1, \Sigma_{11.2})$.

$$A_{12} A_{22}^{-1} A_{21} = -A_{11.2} + A_{11}.$$

A result:
$$\begin{cases} Z \sim \chi^2_{n_1 + n_2} \\ X \sim \chi^2_{n_1} \end{cases}$$

 Then, $Y = (Z - X) \sim \chi^2_{n_2}$

To use the result in our setup, we have to convert A_{11} in such a way that its covariance becomes $\Sigma_{11.2}$.

$$\text{Then, } A_{12} A_{22}^{-1} A_{21} \sim W(\Sigma_{11.2}, p - 1).$$

$$\text{Now, } a_{11.2} \sim \chi^2_{n - p + 1}.$$

$$\therefore \underline{a}_{12} A_{22}^{-1} \underline{a}_{21} \sim \chi^2_{p-1} \quad (\text{After: B.C.U})$$

$$\text{So, } \frac{a_{11.2}}{\sigma_{11.2}} \sim \chi^2_{n-p+1} \quad \begin{matrix} \heartsuit \\ \heartsuit \\ \heartsuit \end{matrix}$$

$$\frac{\underline{a}_{(1)}' A_{22}^{-1} \underline{a}_{(1)}}{\sigma_{11.2}} \sim \chi^2_{p-1}.$$

$$\frac{n-p}{p-1} \cdot \frac{R^2}{1-R^2} = \frac{\chi^2_{p-1} / (p-1)}{\chi^2_{n-p} / (n-p)} \sim F_{p-1, n-p}.$$